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Electrosynthesis of ethylene from syngas

In the format provided by the authors and unedited

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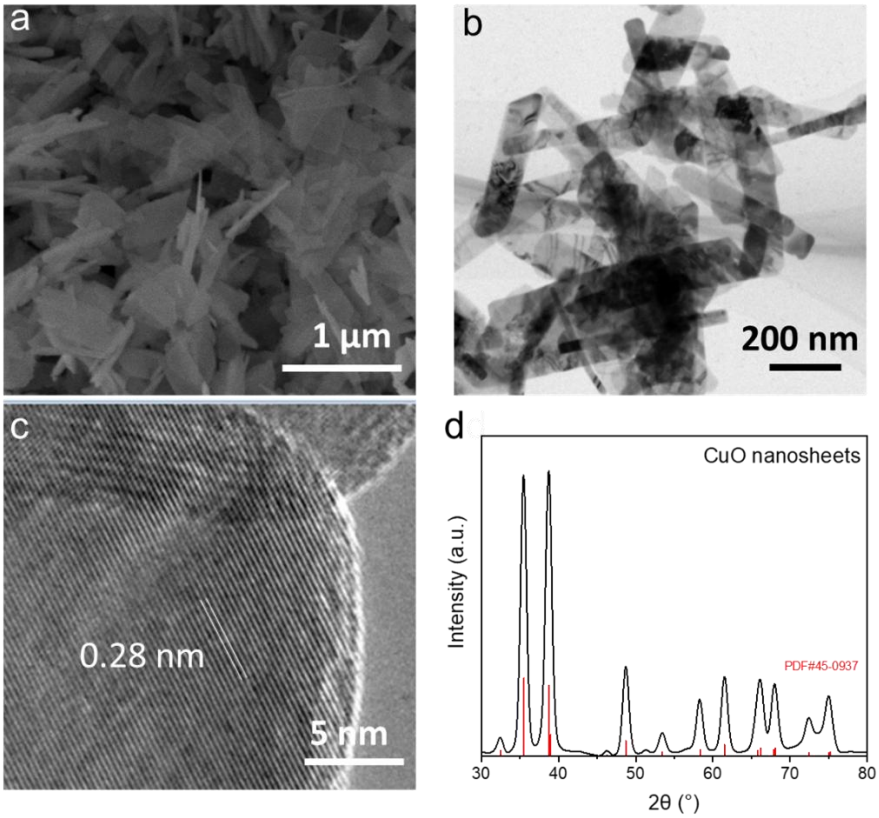
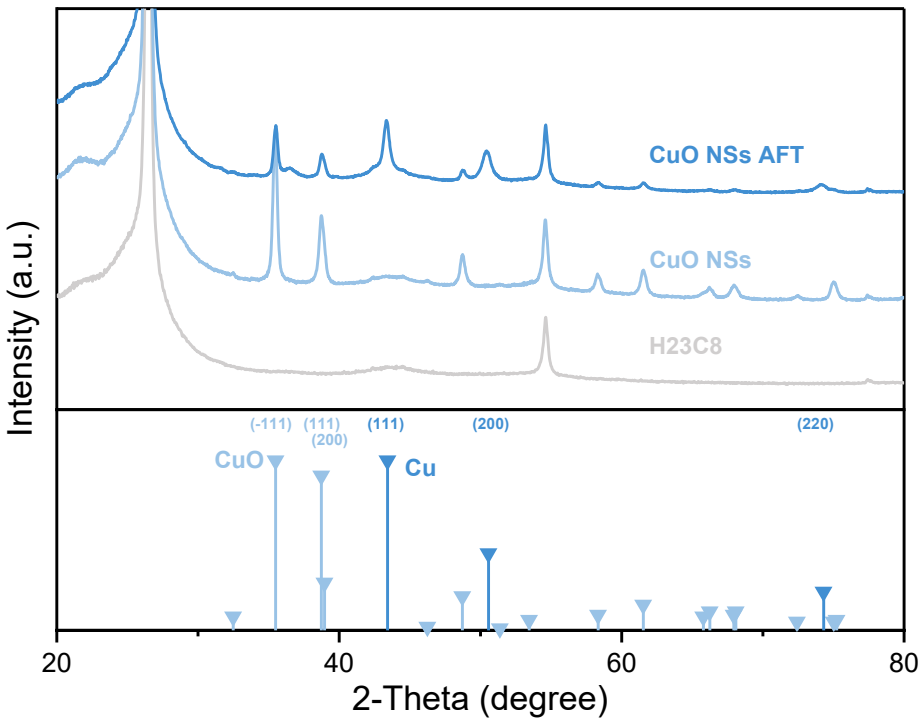
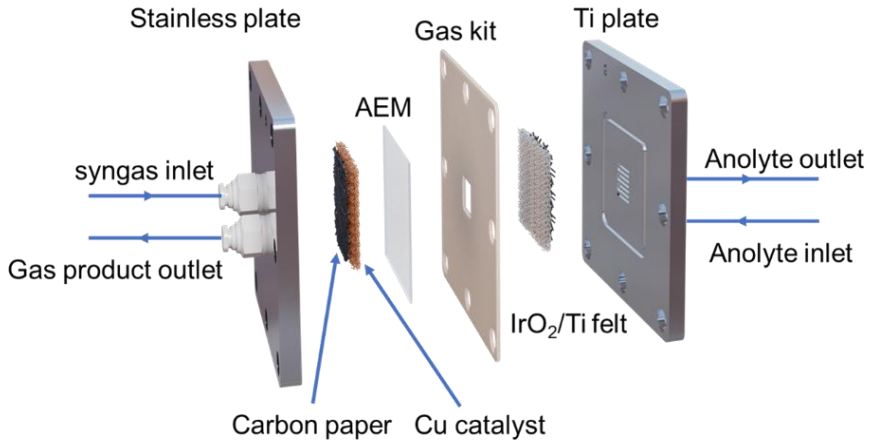


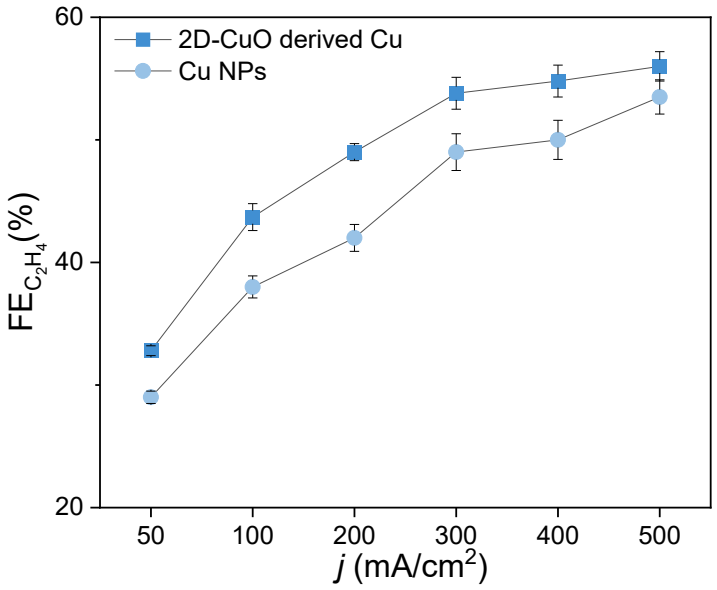
Supplementary Fig. 1 | Characterization of CuO nanosheet. a,b, SEM (a) and TEM (b) images of pristine CuO nanosheet. **c,** HRTEM image of CuO nanosheets, which demonstrates the CuO nanosheets are well crystallized with a periodic fringe of 0.28 nm, according well to the d-spacing value of the (110) plane for monoclinic CuO. d, XRD pattern of CuO nanosheets. The pattern of CuO (PDF#45-0937) was used as a reference.



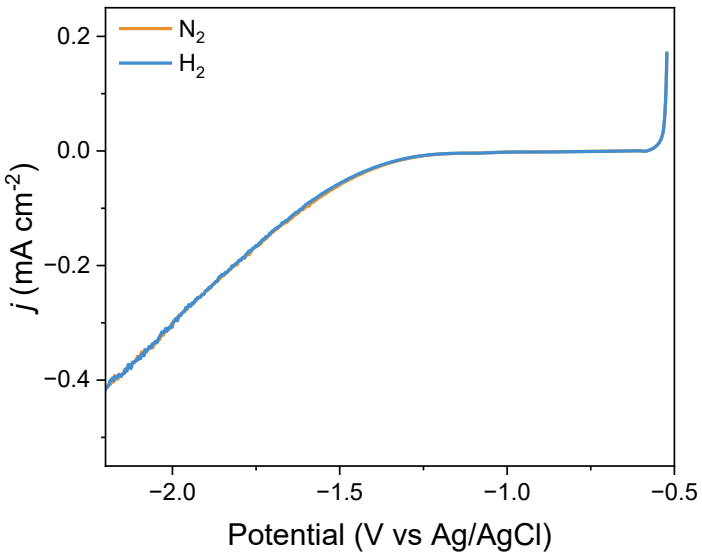
Supplementary Fig. 2 | XRD patterns. XRD pattern of H23C3 carbon paper, CuO nanosheets before (CuO NSs) and after (CuO NSs AFT) COR. XRD patterns of Cu (PDF#04-0836) and CuO (PDF#45-0937) were used as the reference. The new peaks located at 43.3˚, 50.4˚, and 74.1˚ were indexed to the (111), (200), and (220) facets of cubic Cu, respectively.



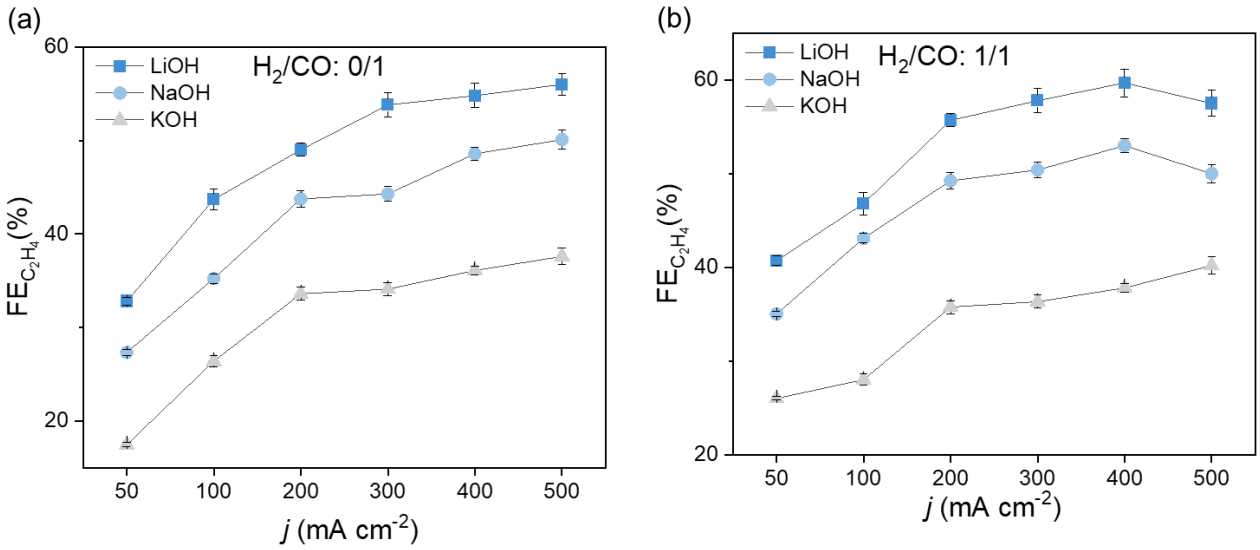
Supplementary Fig. 3 | Schematic illustration of zero-gap membrane electrode assembly (MEA) electrolyzer.



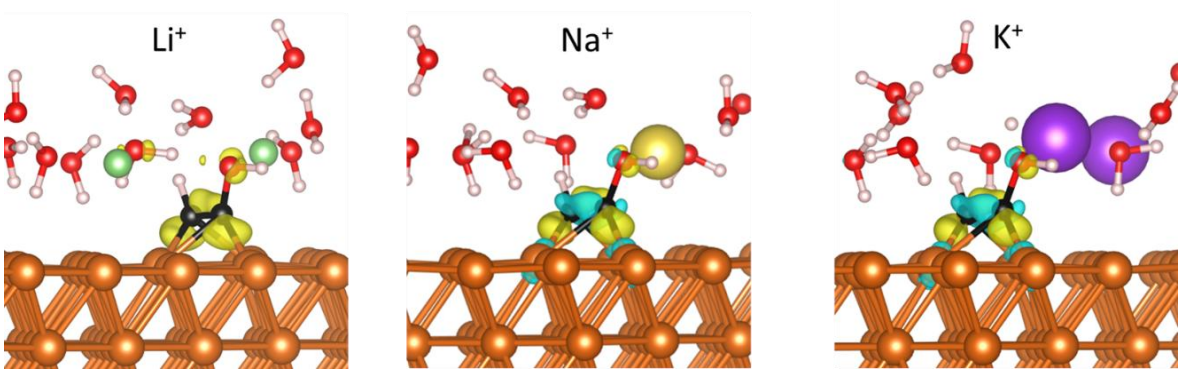
Supplementary Fig. 4 | Performance of COR over commercial Cu nanoparticle and 2D-CuO-derived Cu at flow rate of 40 sccm. Where error bars are shown, values are means, and error bars indicate s.d. (n = 3 replicates).



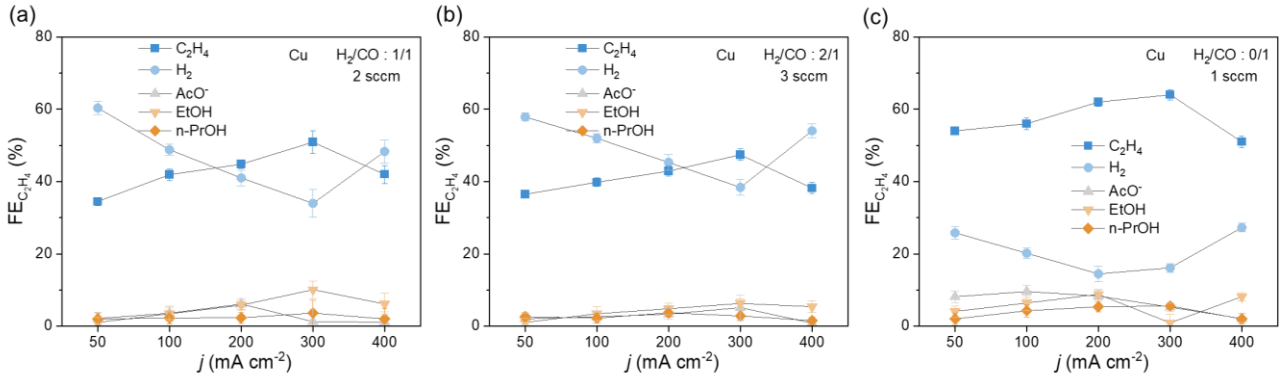
Supplementary Fig. 5 | LSV curves of Cu under N_2 and H_2 conditions.



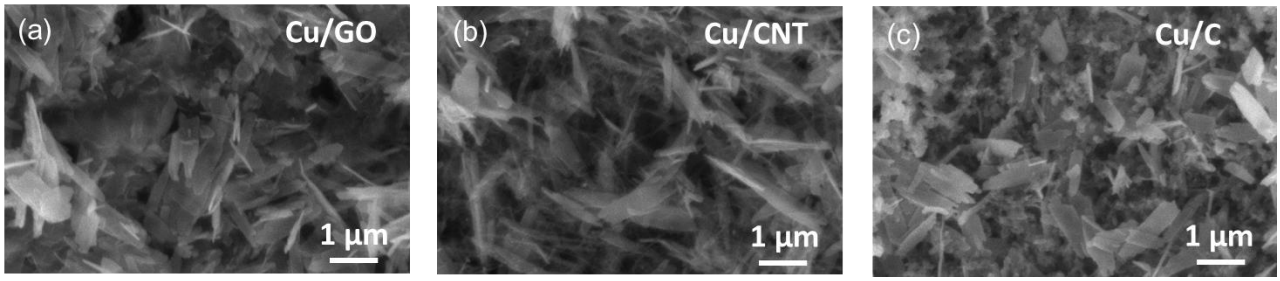
Supplementary Fig. 6 | Influence of electrolyte on C_2H_4 selectivity. a,b, Faradic efficiency of C_2H_4 during the COR from (a) pure CO and (b) H_2/CO ratio of 1/1 at 40 sccm in different electrolytes. Where error bars are shown, values are means, and error bars indicate s.d. (n = 3 replicates).



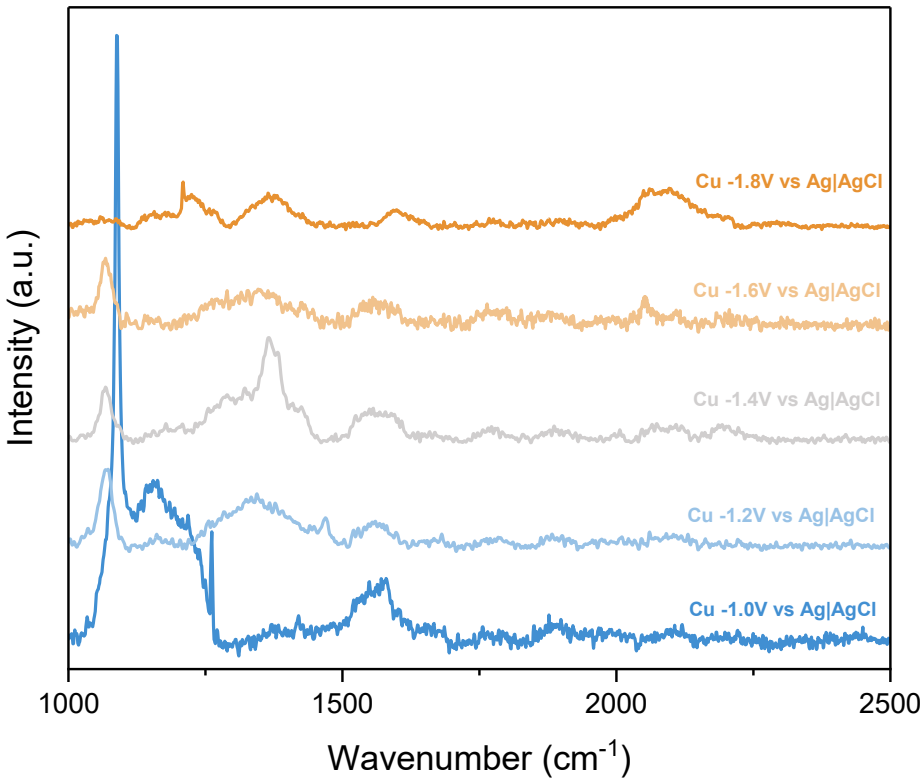
Supplementary Fig. 7 | Interactions between HCCOH* intermediate and various cations on Cu surface. Charge density difference plot for HCCOH* intermediate on Cu surfaces with Li^+, Na^+ and K^+, respectively. Yellow and blue contours represent the iso-surfaces of electronic charge accumulation and depletion, respectively, with an iso-surface value of 0.01 e $Å^{-3}$ implemented.



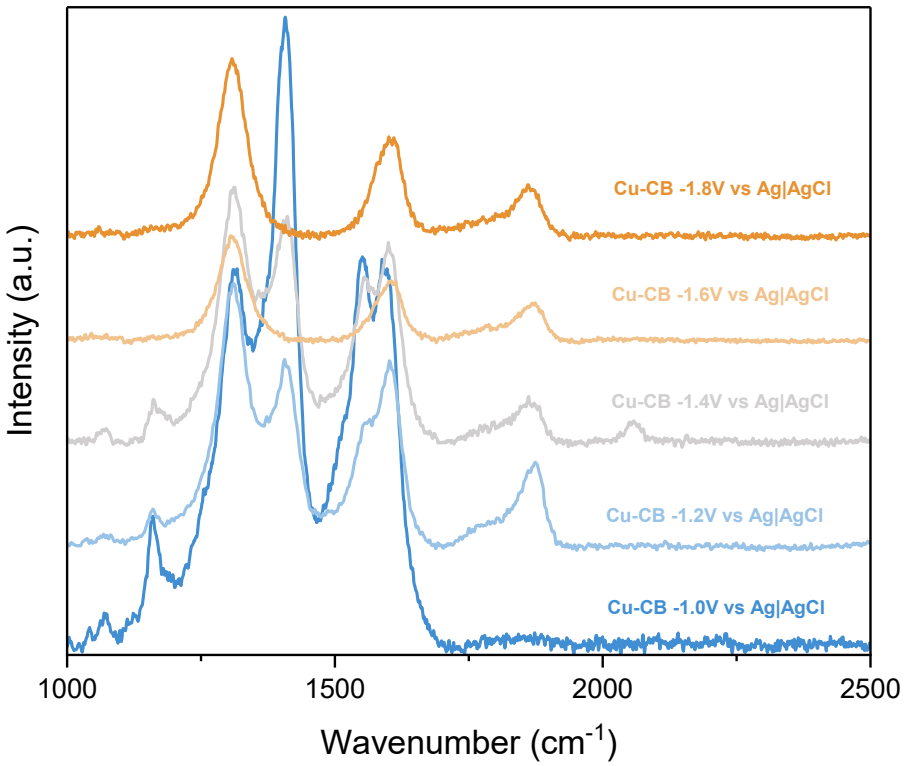
Supplementary Fig. 8 | FE of all products under different applied current density for COR over pristine Cu from syngas and pure CO. a, b of H_2/CO ratio 1/1 at 2 sccm (a) and 2/1 at 3 sccm (b). c, pure CO at 1sccm. Note H_2 FE for COR from syngas was estimated based on the FE of other products. Where error bars are shown, values are means, and error bars indicate s.d. (n = 3 replicates).



Supplementary Fig. 9 | Characterization of the Cu electrode modified by various carbon materials. a,b,c, SEM images of Cu electrode with (a) GO, (b) CNT, and (c) C on carbon paper.



Supplementary Fig. 10 | Operando Raman test. Operando Raman test of pristine Cu under the potentials from –1.0 V to –1.8 V (vs. Ag/AgCl reference electrode filled with saturated aqueous KCl) in electrolytes of 1 M LiOH. The CO adsorption onset potential is –1.6 V.



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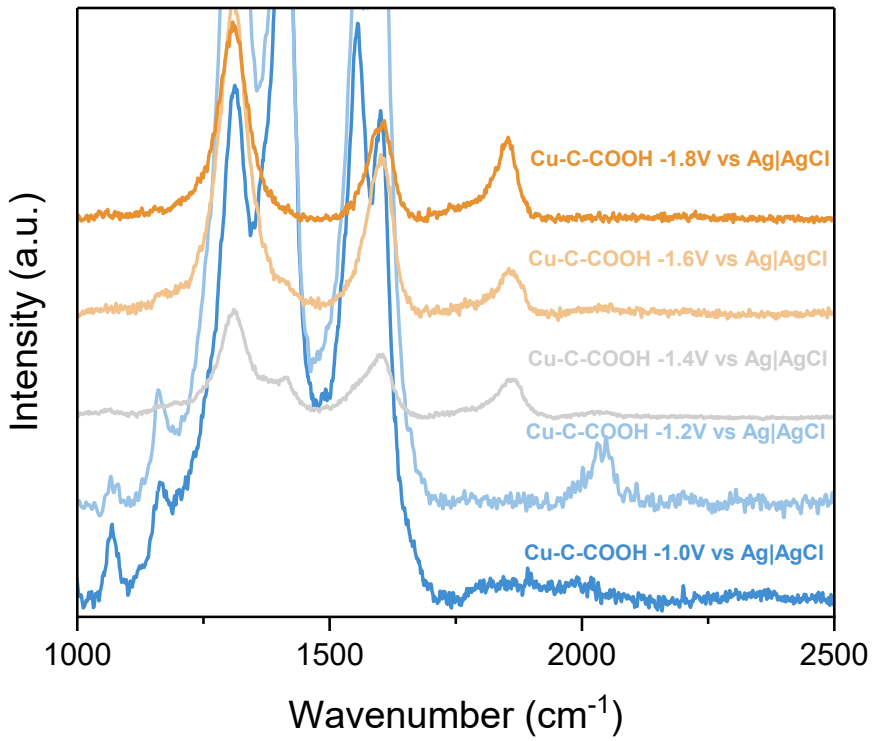

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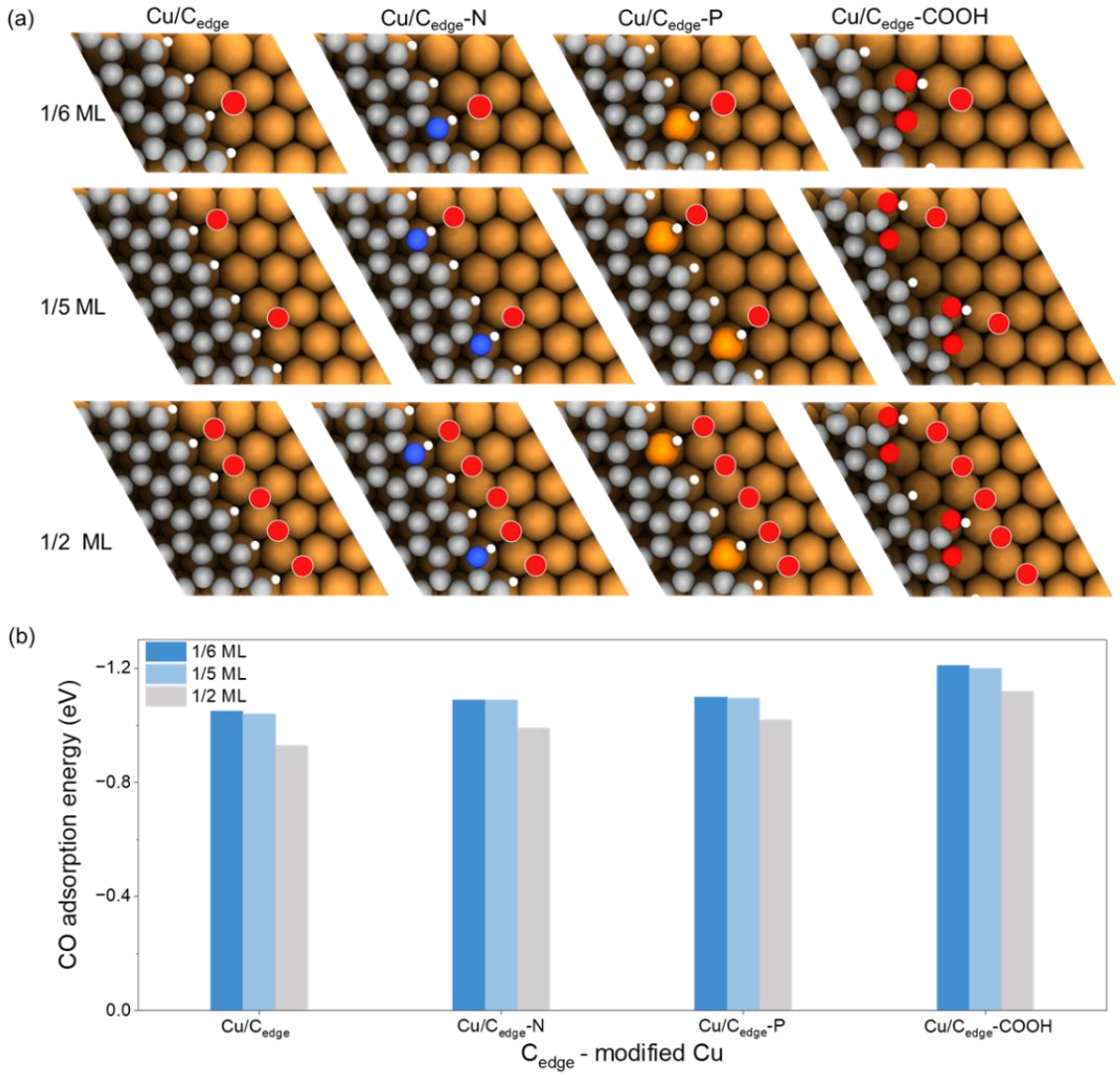
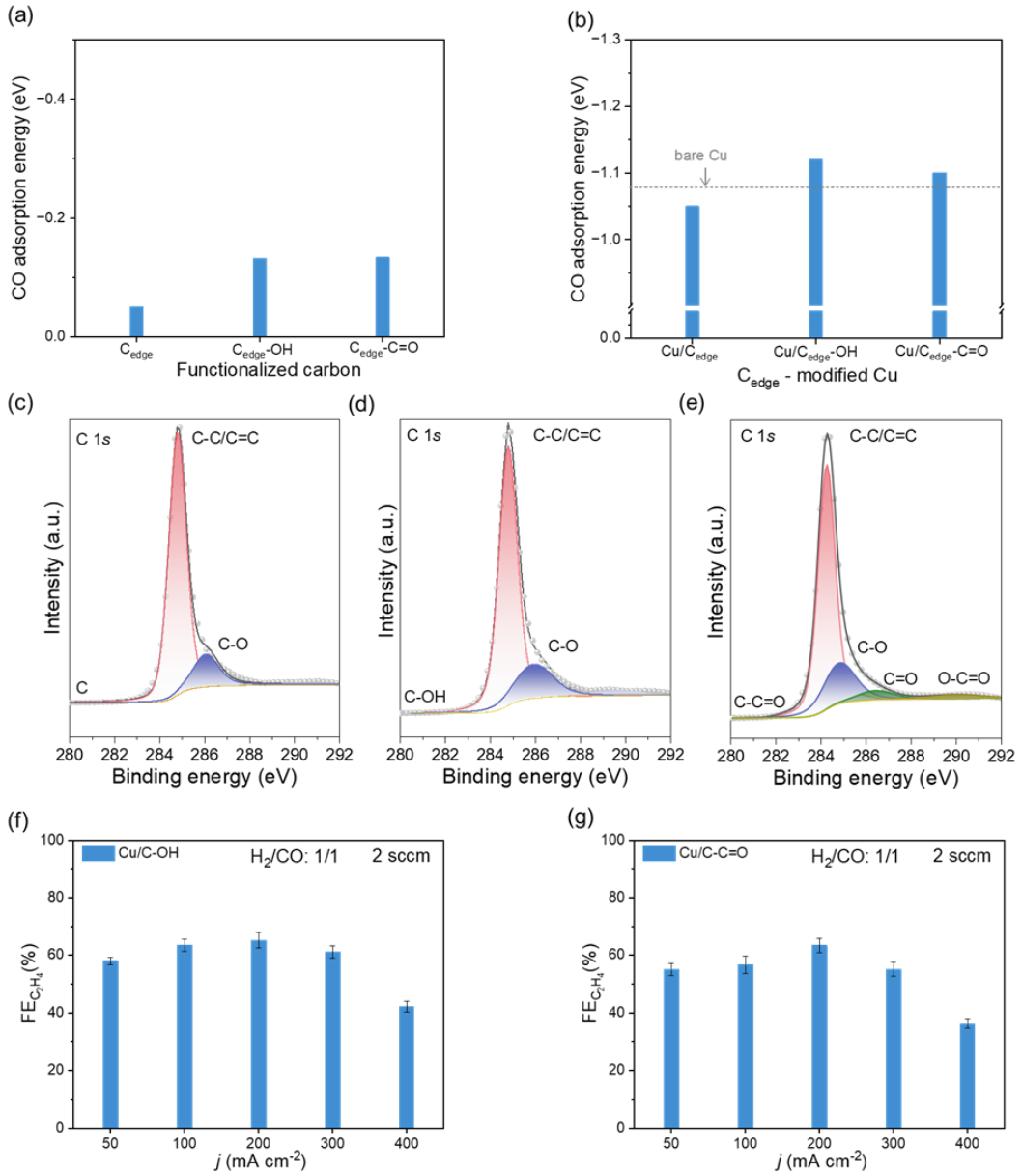


Supplementary Fig. 13 | CO coverage effect on CO adsorption energy on carbon. a,b, (a) Geomertries and (b) adsorption energies of CO adsorbed on C-modified Cu surfaces at coverages of 1/6 ML 1/5 ML, and 1/2 ML.



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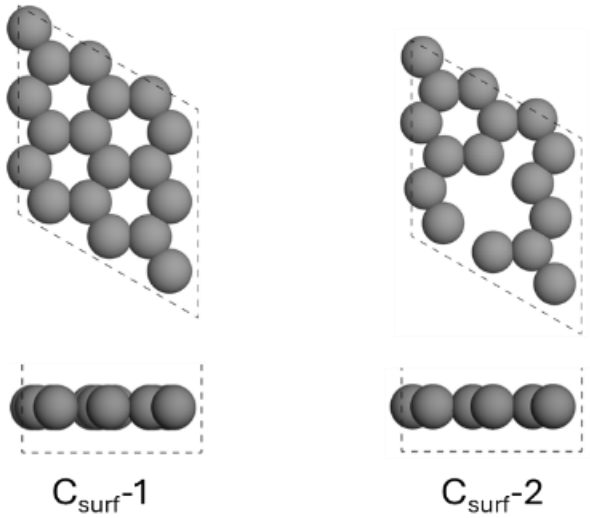
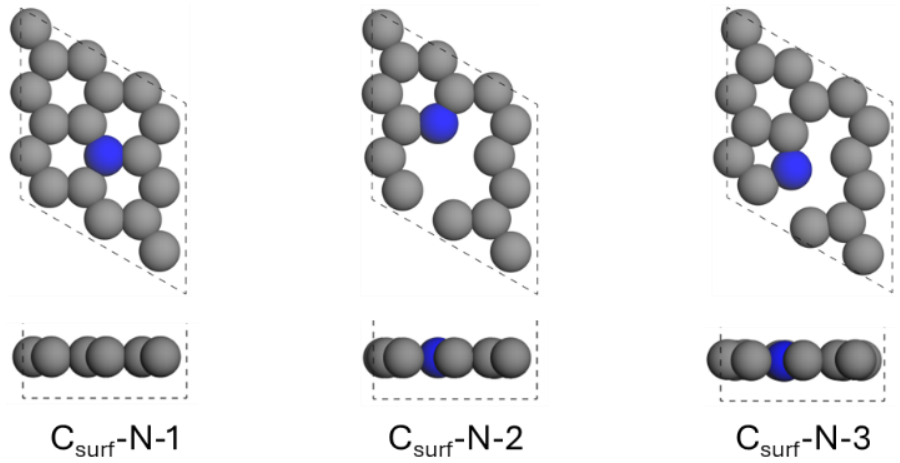
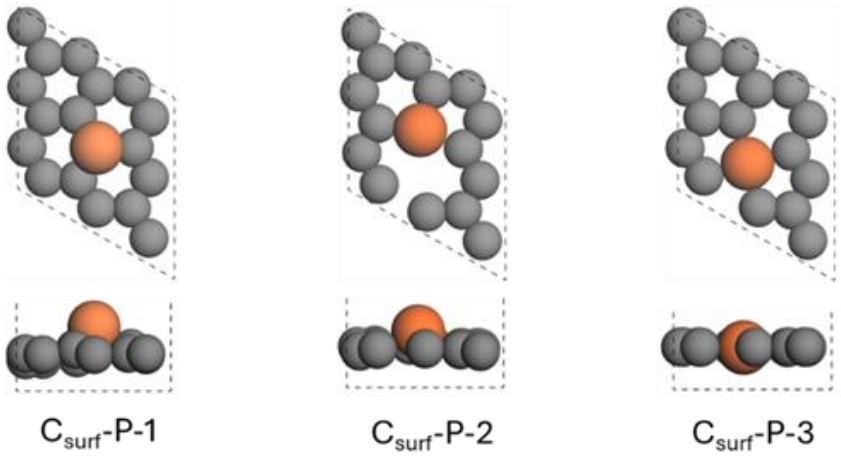


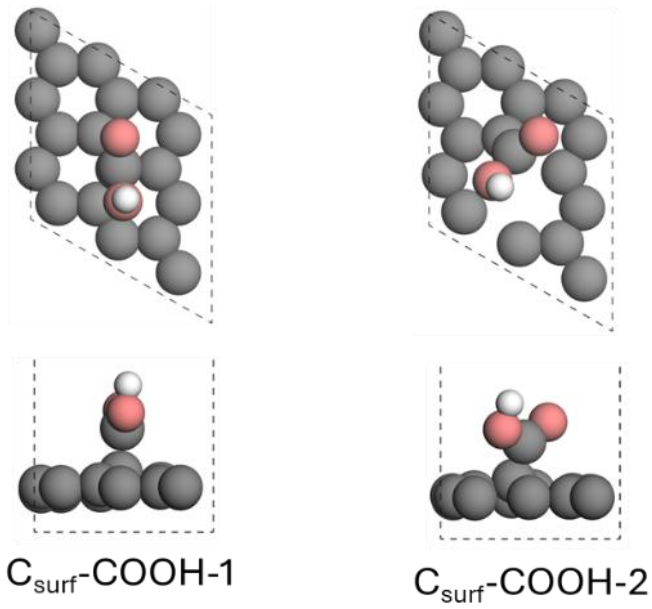
Supplementary Fig. 15 | Optimized geometries of carbon surfaces.



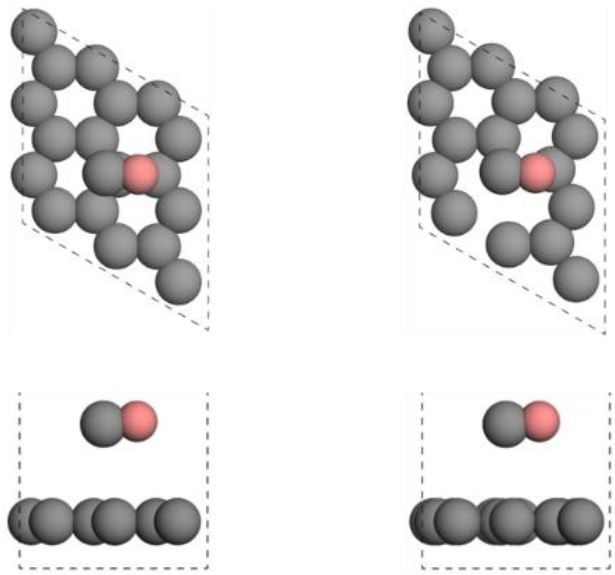
Supplementary Fig. 16 | Optimized geometries of N-functionalized carbon surfaces.



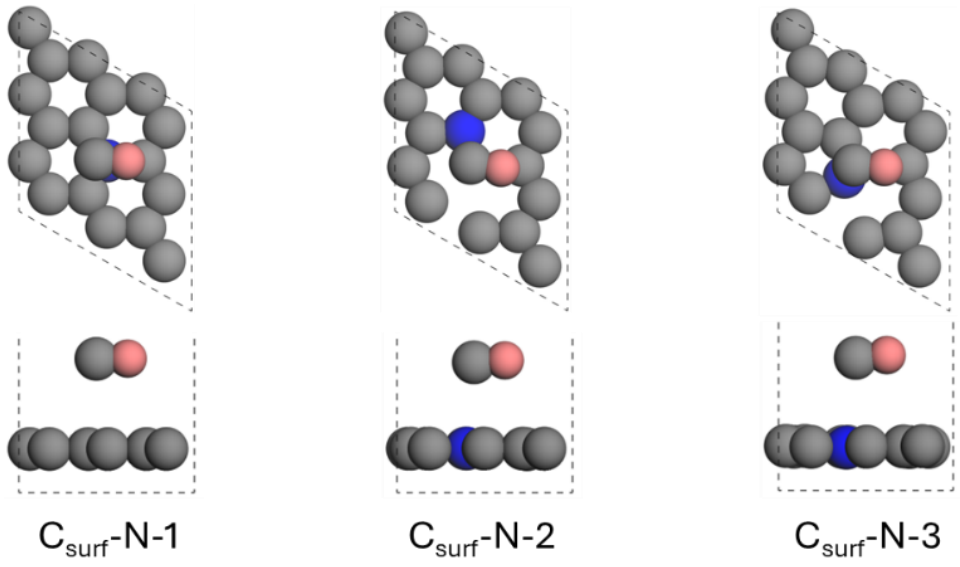
Supplementary Fig. 17 | Optimized geometries of P-functionalized carbon surfaces.



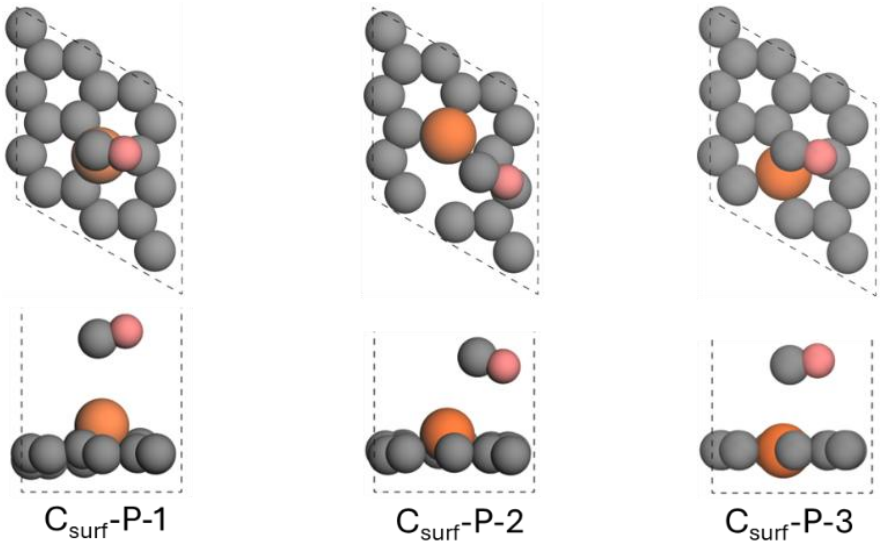
Supplementary Fig. 18 | Optimized geometries of COOH-functionalized carbon surfaces.



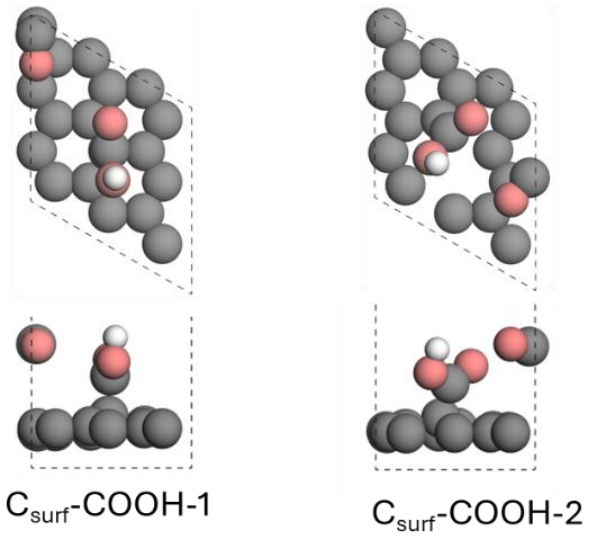
Supplementary Fig. 19 | Optimized geometries of CO adsorption on carbon surfaces.



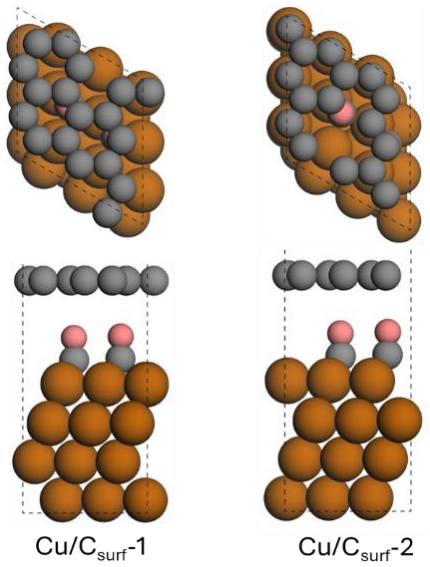
Supplementary Fig. 20 | Optimized geometries of CO adsorption on N-functionalized carbon surfaces.



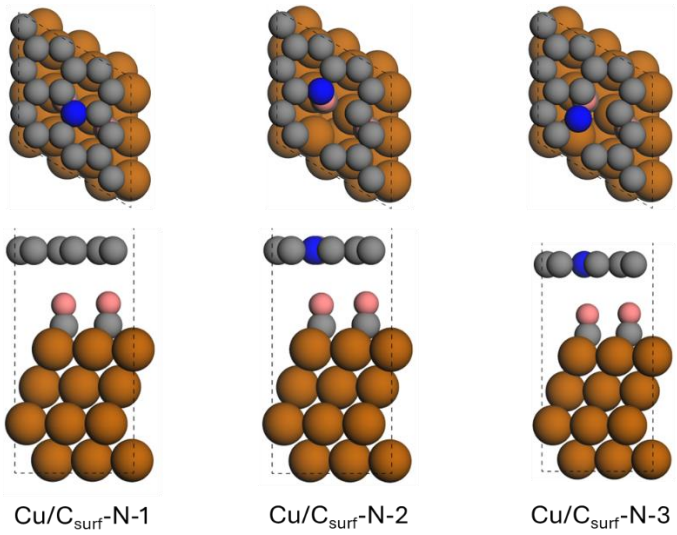
Supplementary Fig. 21 | Optimized geometries of CO adsorption on P-functionalized carbon surfaces.



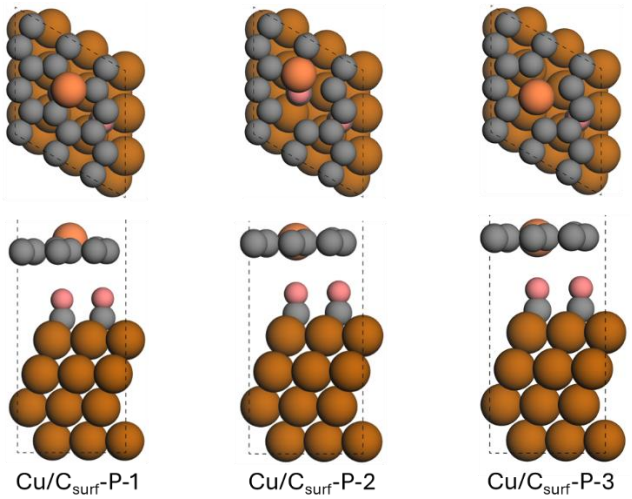
Supplementary Fig. 22 | Optimized geometries of CO adsorption on COOH-functionalized carbon surfaces.



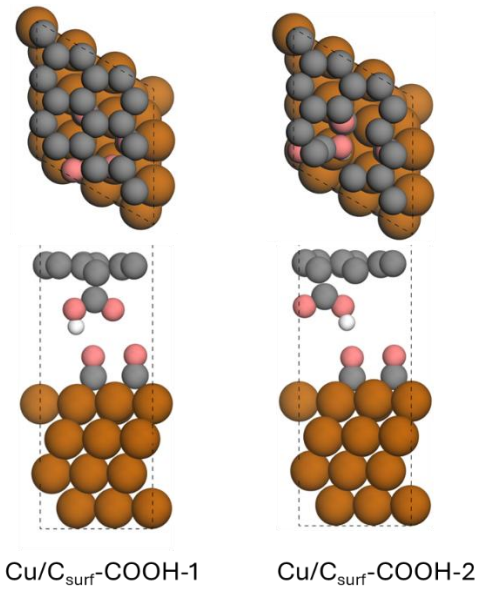
Supplementary Fig. 23 | Optimized geometries of CO adsorption on Cu modified with carbon surfaces.



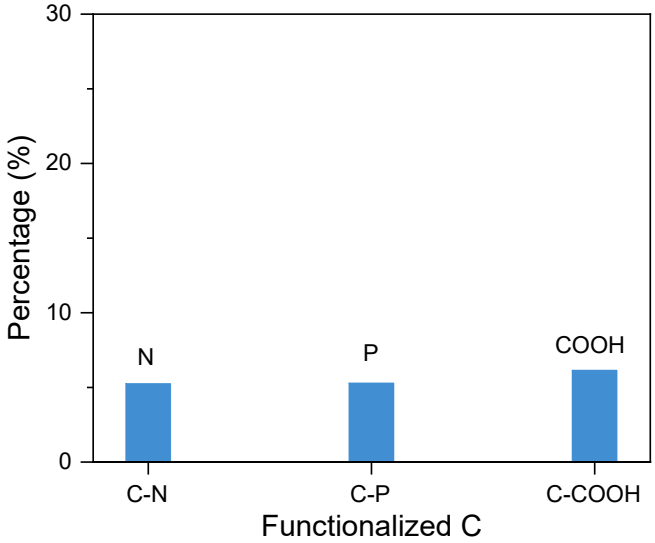
Supplementary Fig. 24 | Optimized geometries of CO adsorption on Cu modified with N-functionalized carbon surfaces.



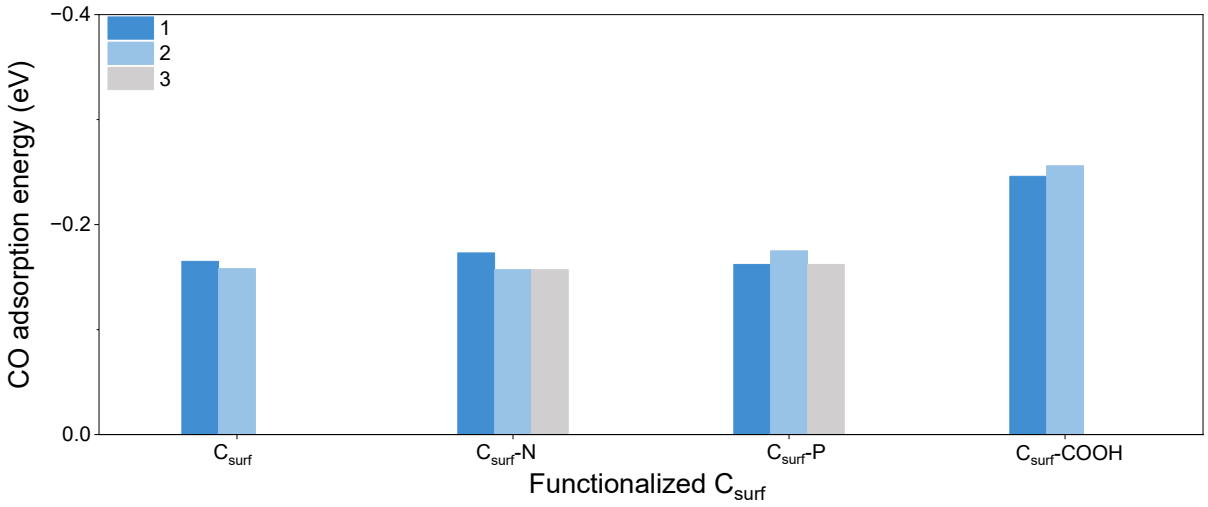
Supplementary Fig. 25 | Optimized geometries of CO adsorption on Cu modified with P-functionalized carbon surfaces.



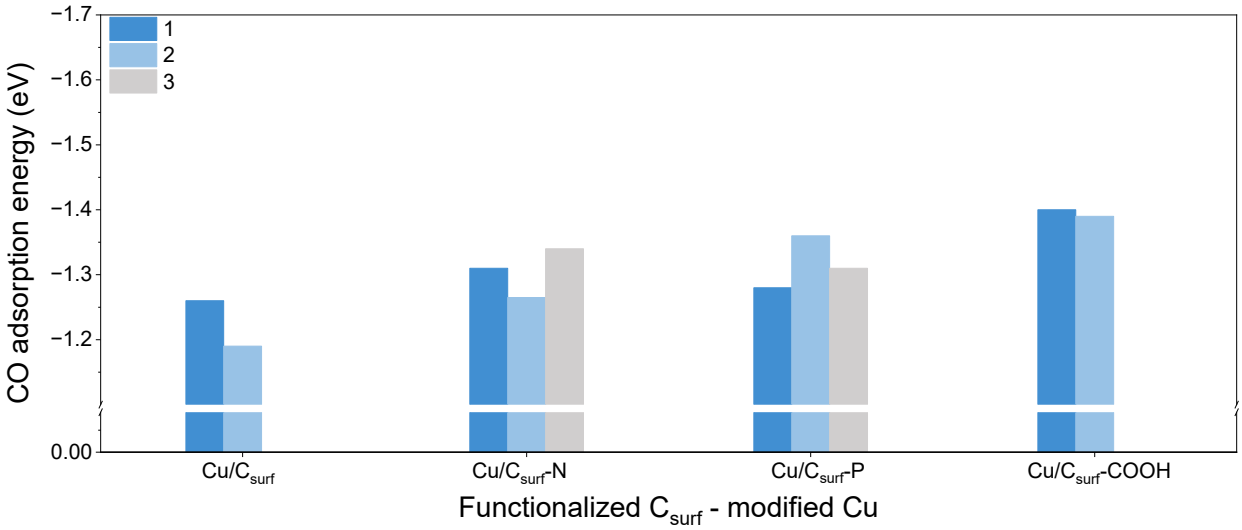
Supplementary Fig. 26 | Optimized geometries of CO adsorption on Cu modified with COOH-functionalized carbon surfaces.



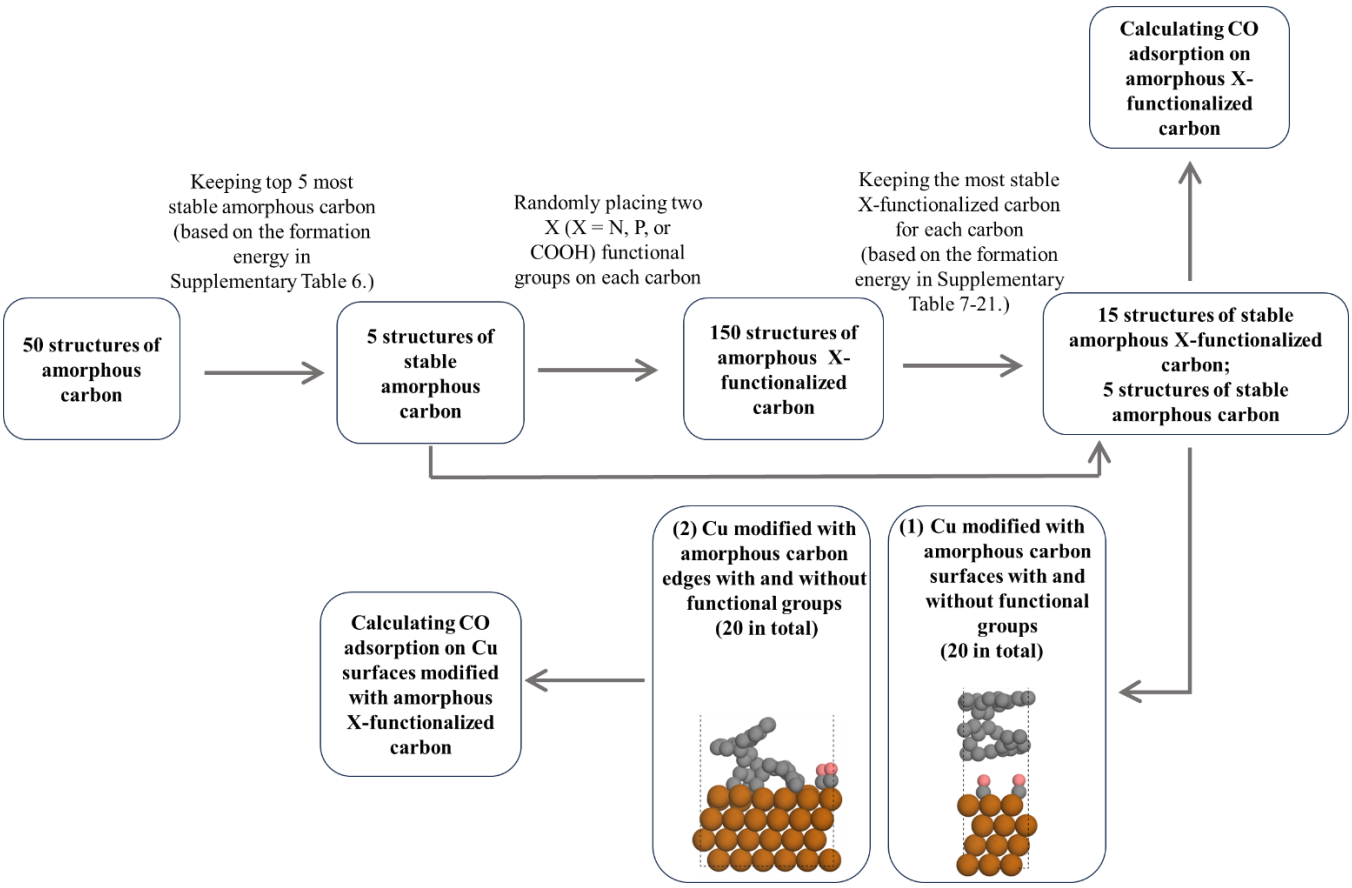
Supplementary Fig. 27 | Percentage of functional group in carbon materials, derived from XPS C 1*s* analysis.



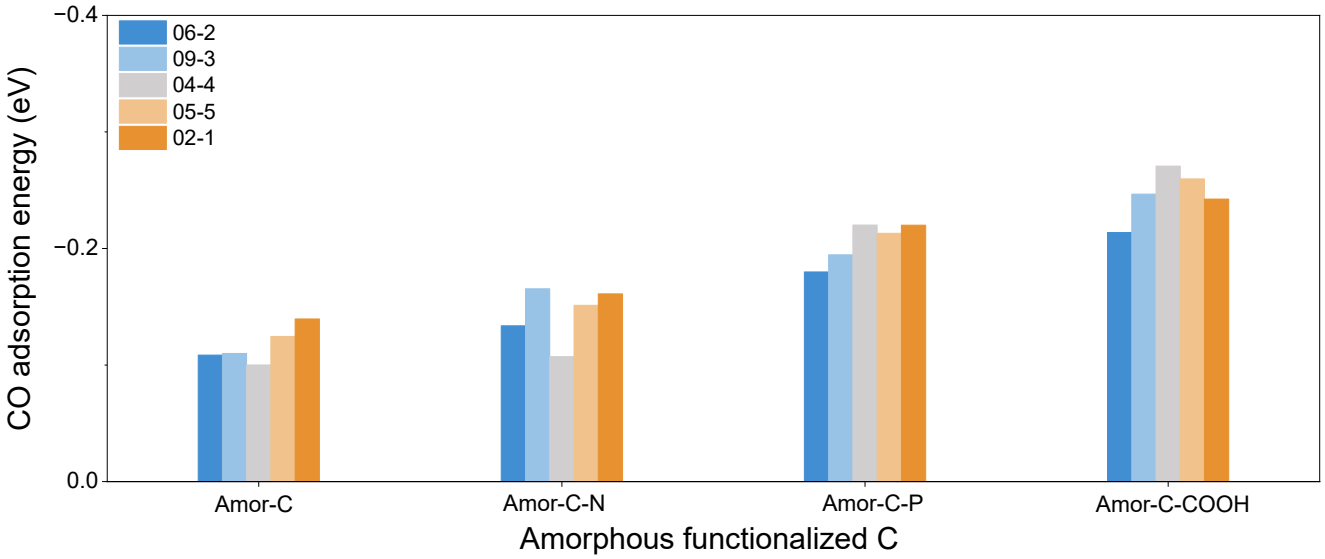
Supplementary Fig. 28 | CO adsorption energies on functionalized carbon surfaces.



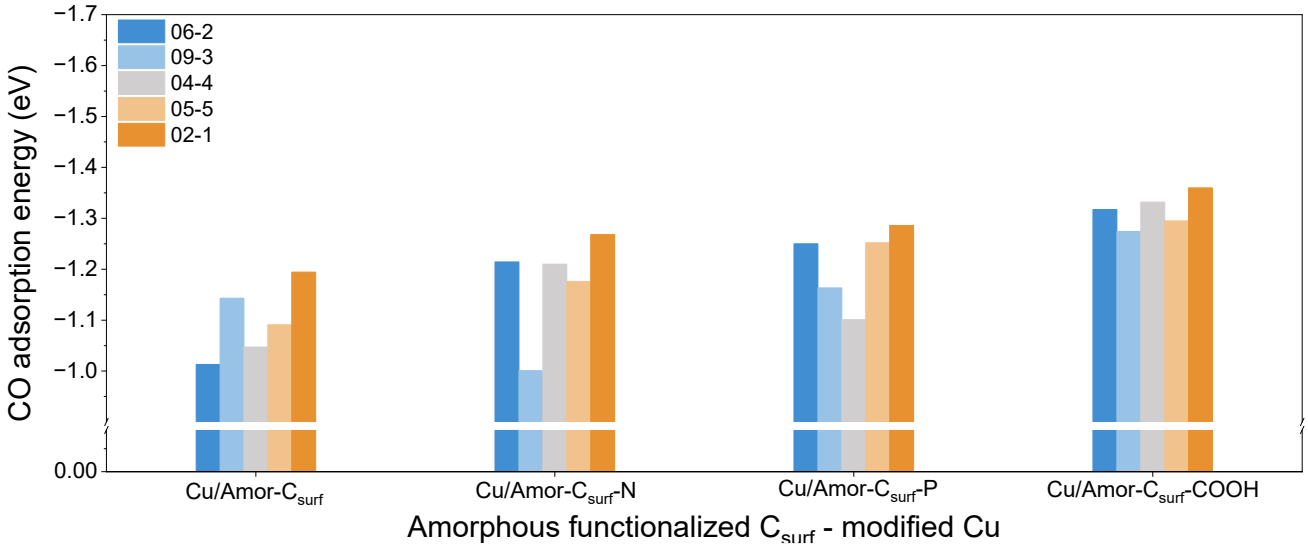
Supplementary Fig. 29 | CO adsorption energies on Cu modified with functionalized carbon surfaces.



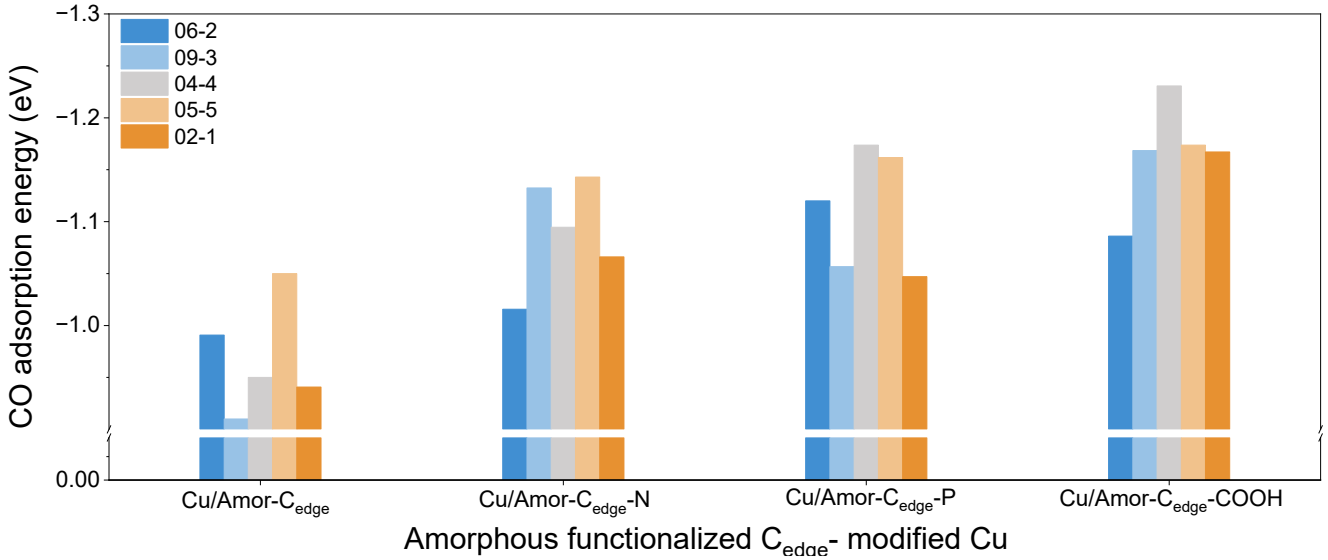
Supplementary Fig. 30 | Pipeline for the generation of amorphous X-functionalized carbon structures (X = N, P, and COOH) and modified Cu surfaces.



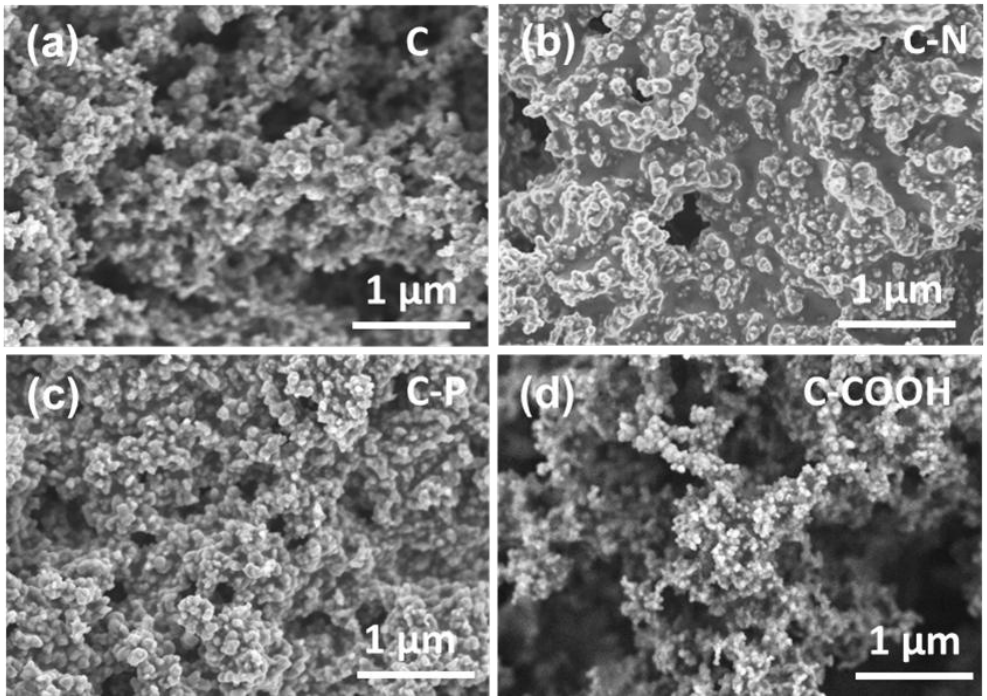
Supplementary Fig. 31 | CO adsorption energies on the most stable amorphous X-functionalized C (X = N, P, and COOH) in Supplementary Table 7 - 21.



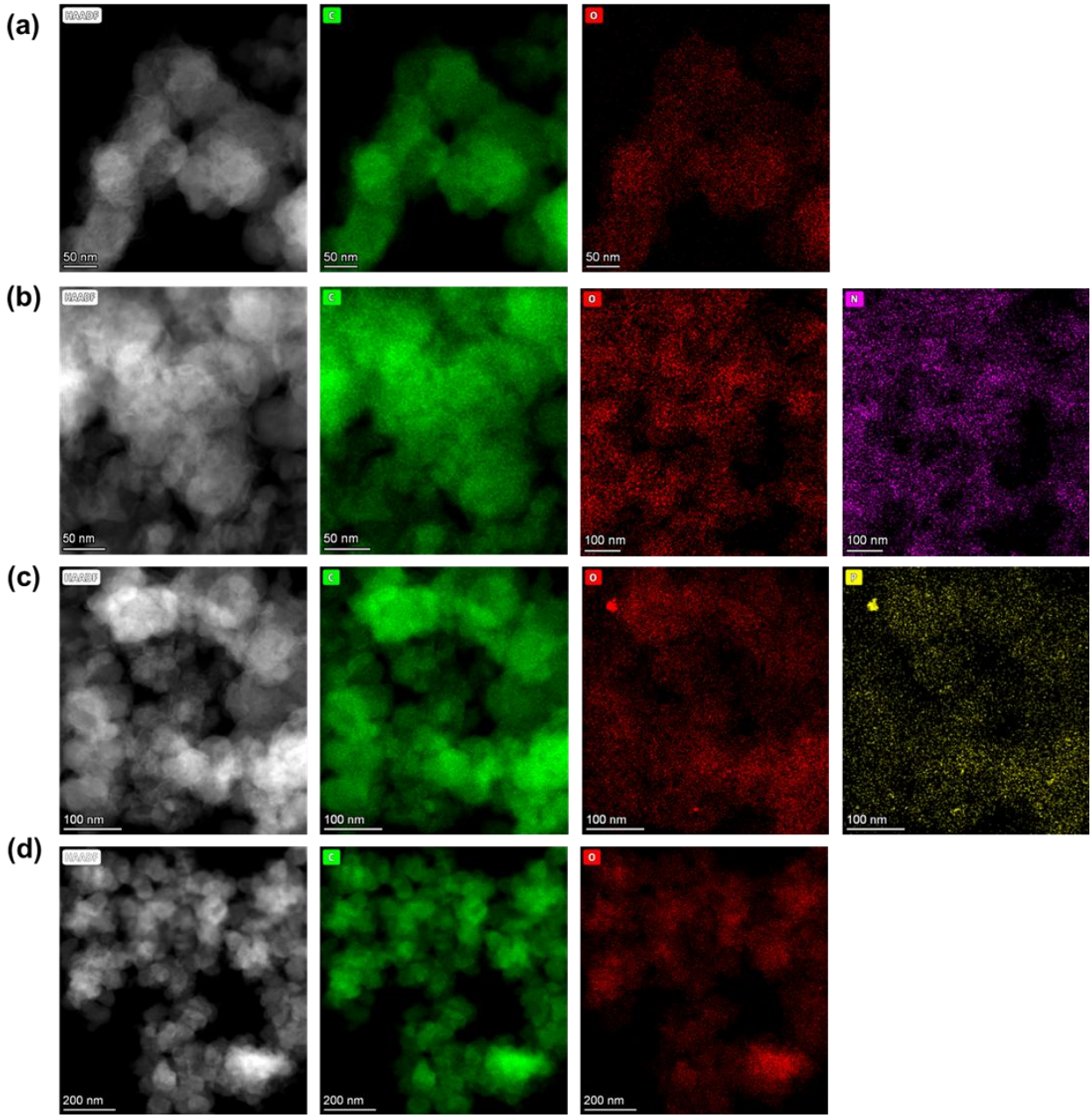
Supplementary Fig. 32 | CO adsorption energies on the Cu modified with the most stable amorphous X-functionalized C surfaces (X = N, P, and COOH) in Supplementary Table 7-21.



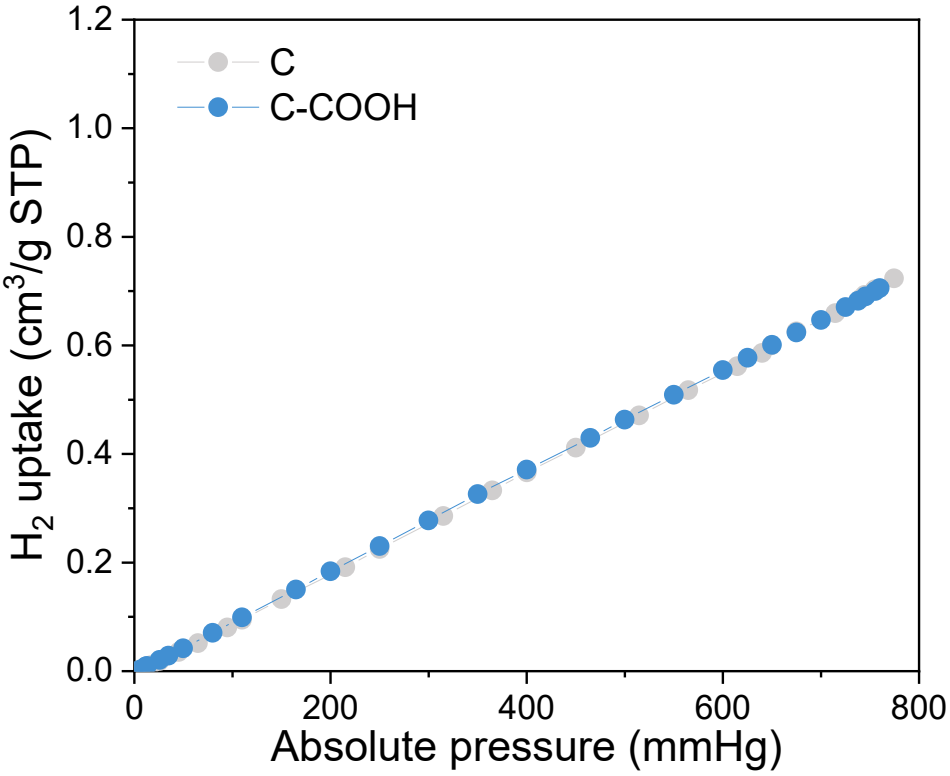
Supplementary Fig. 33 | CO adsorption energies on the Cu modified with amorphous X-functionalized C edges (X = N, P, and COOH).



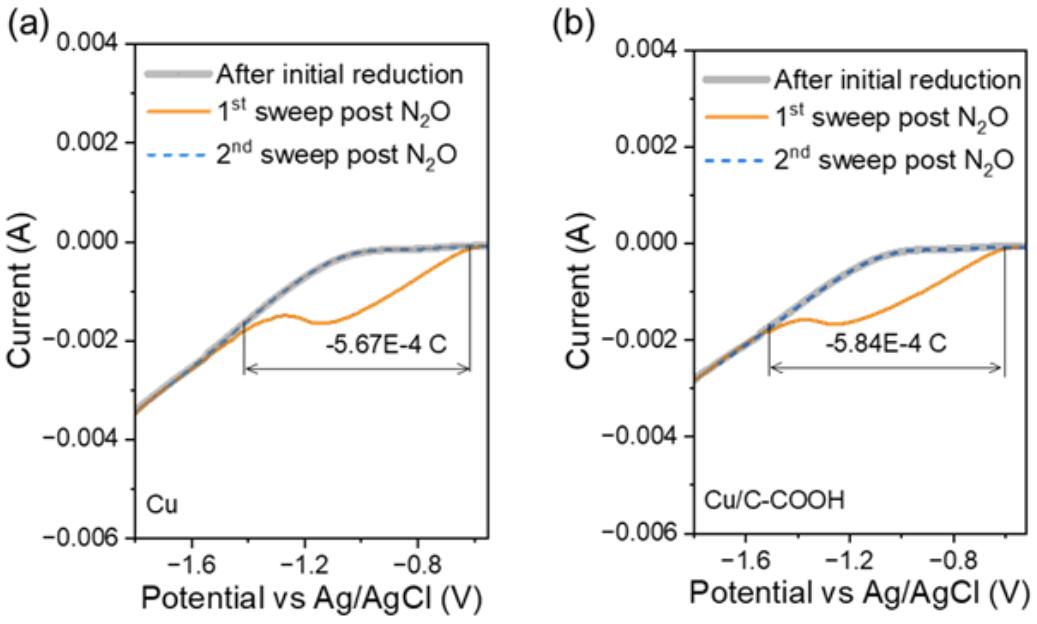
Supplementary Fig. 34 | Characterization of various carbon materials. a,b,c, SEM images of (a) C, (b) C–N, (c) C–P and (d) C–COOH.



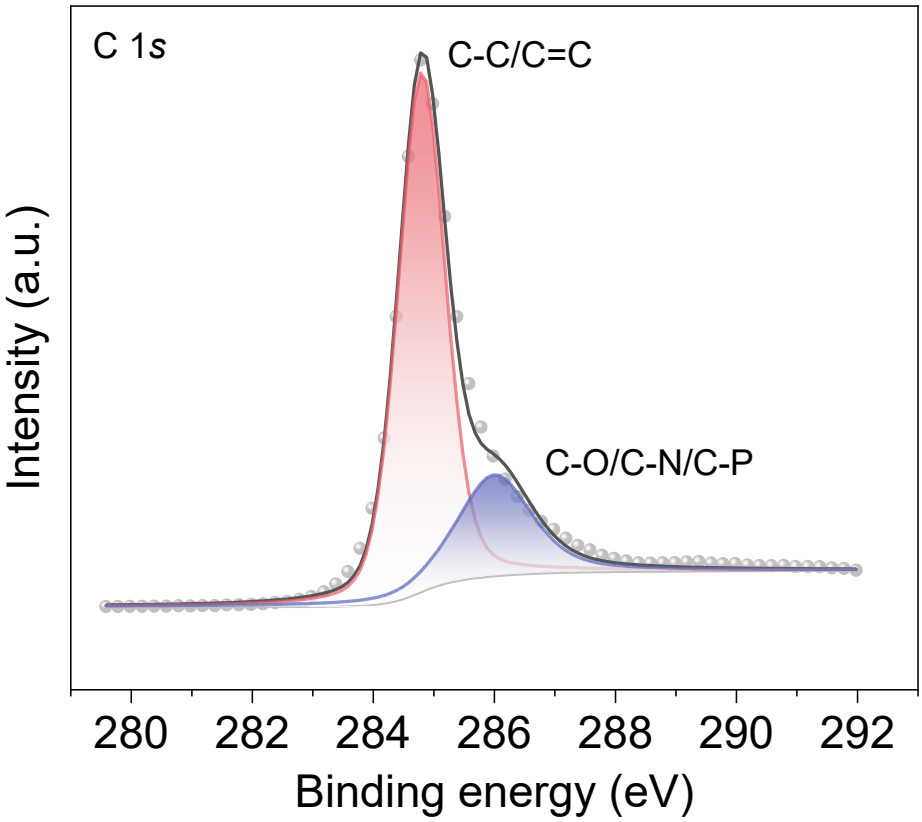
Supplementary Fig. 35 | HAAADF-STEM images and EDS mapping. **a,** C. **b,** C–N. **c,** C–P. **d,** C–COOH.



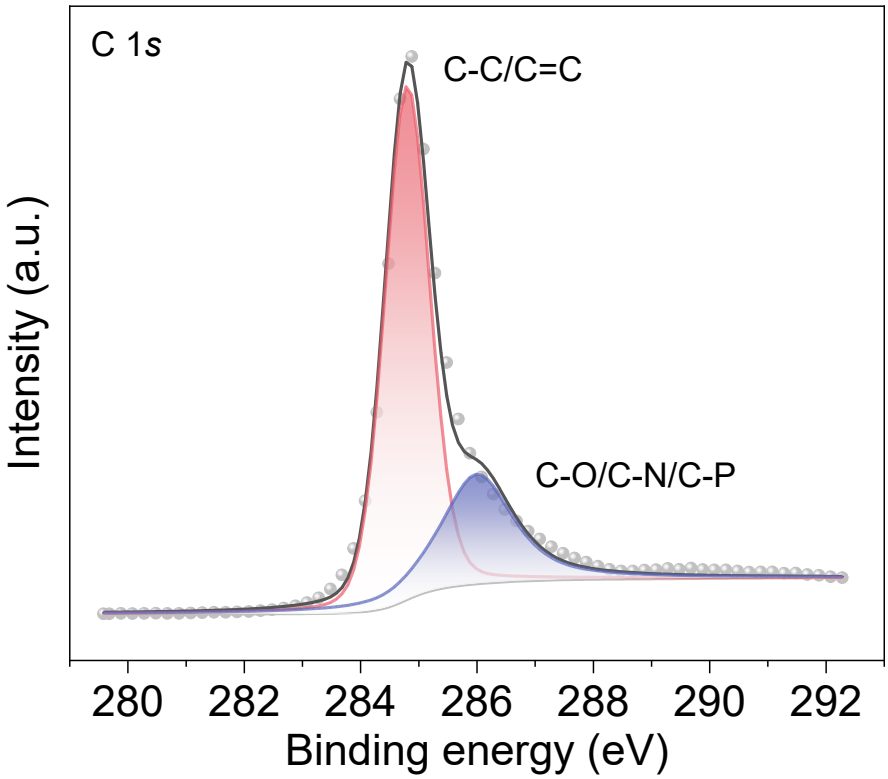
Supplementary Fig. 36 | Adsorption isotherms of H_2 for C and C–COOH at 298 K.



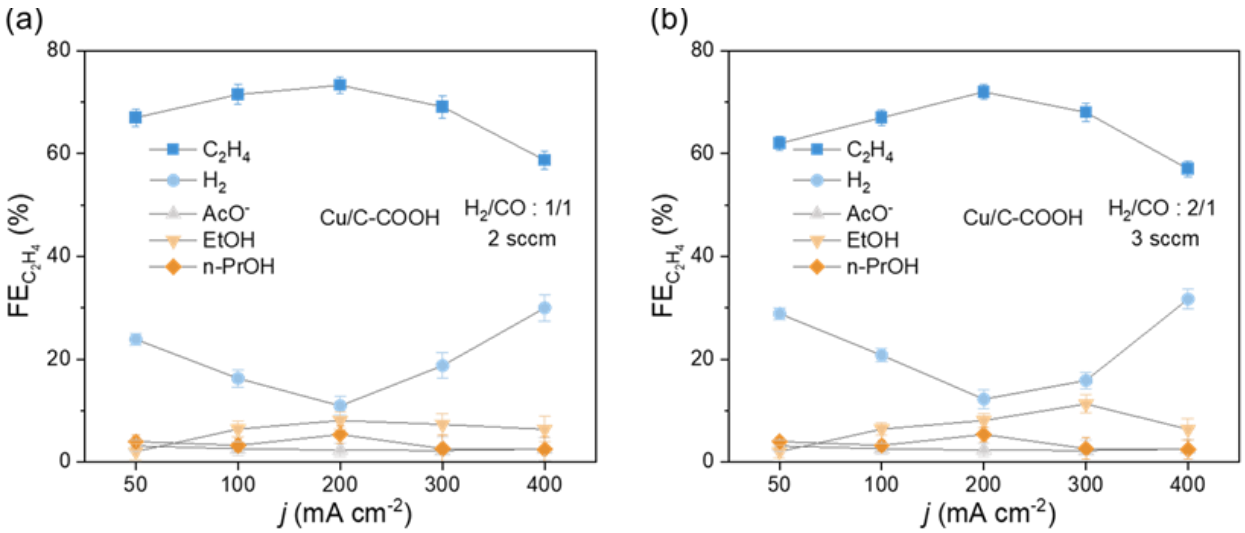
Supplementary Fig. 37 | Coulometric reduction of (a) Cu and (b) Cu/C–COOH electrodes after exposure to N_2O.



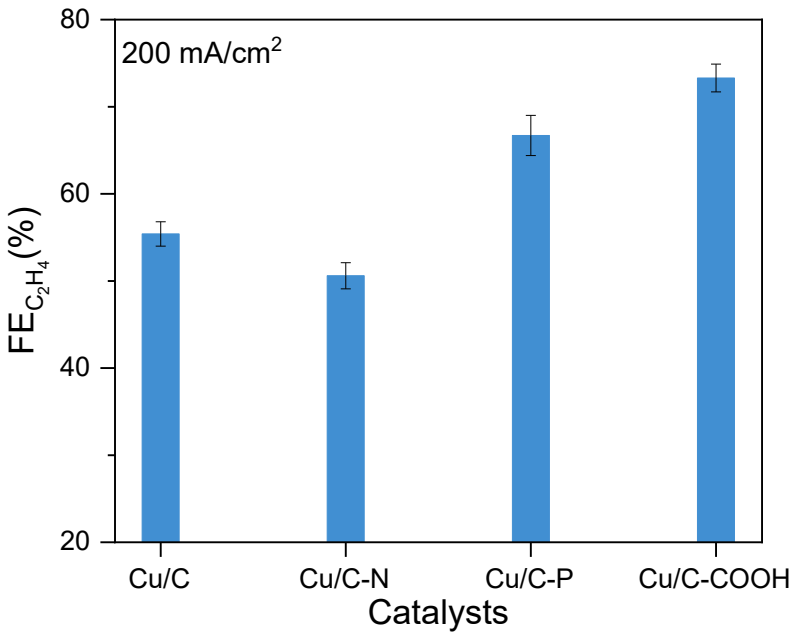
Supplementary Fig. 38 | XPS spectra. XPS C 1*s* spectra for C–N.



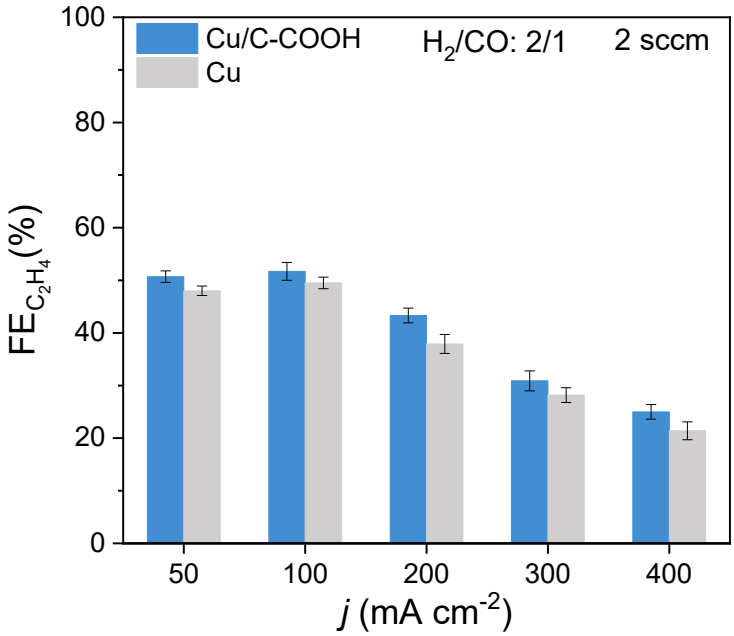
Supplementary Fig. 39 | XPS Spectra. XPS C 1*s* spectra for C–P.



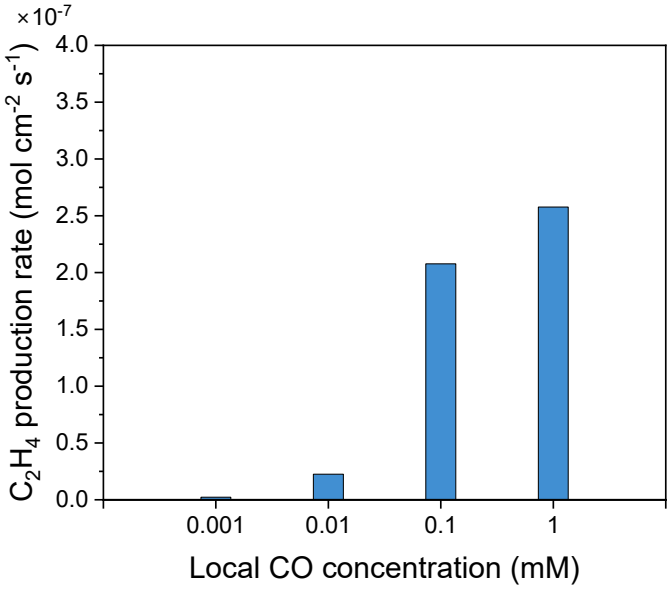
Supplementary Fig. 40 | FE of all products under different applied current density for COR over pristine Cu/C-COOH from syngas. **a, b** H_2/CO ratio of 1/1 at 2 sccm (a) and 2/1 at 3 sccm (b). Where error bars are shown, values are means, and error bars indicate s.d. (n = 3 replicates).



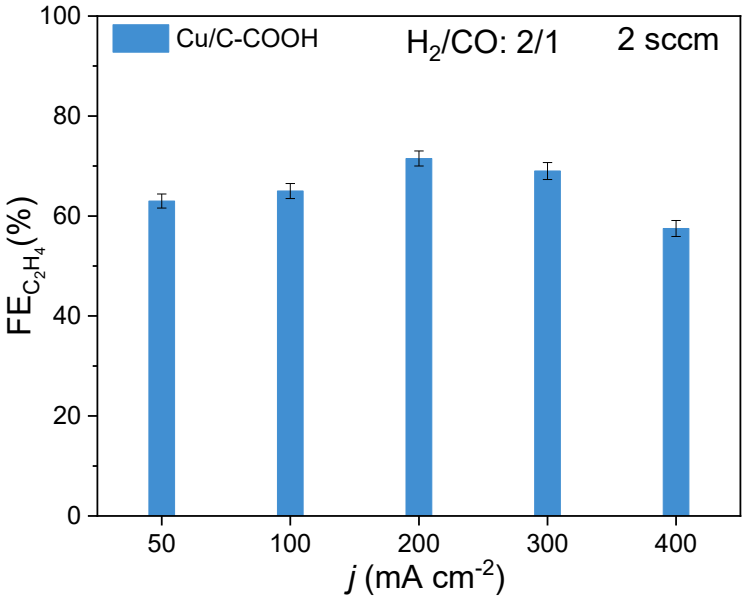
Supplementary Fig. 41 | Effect of functional group on ethylene selectivity. Ethylene FE of COR from 1/1-syngas on Cu modified by C, C–N, C–P and C–COOH. Where error bars are shown, values are means, and error bars indicate s.d. (n = 3 replicates).



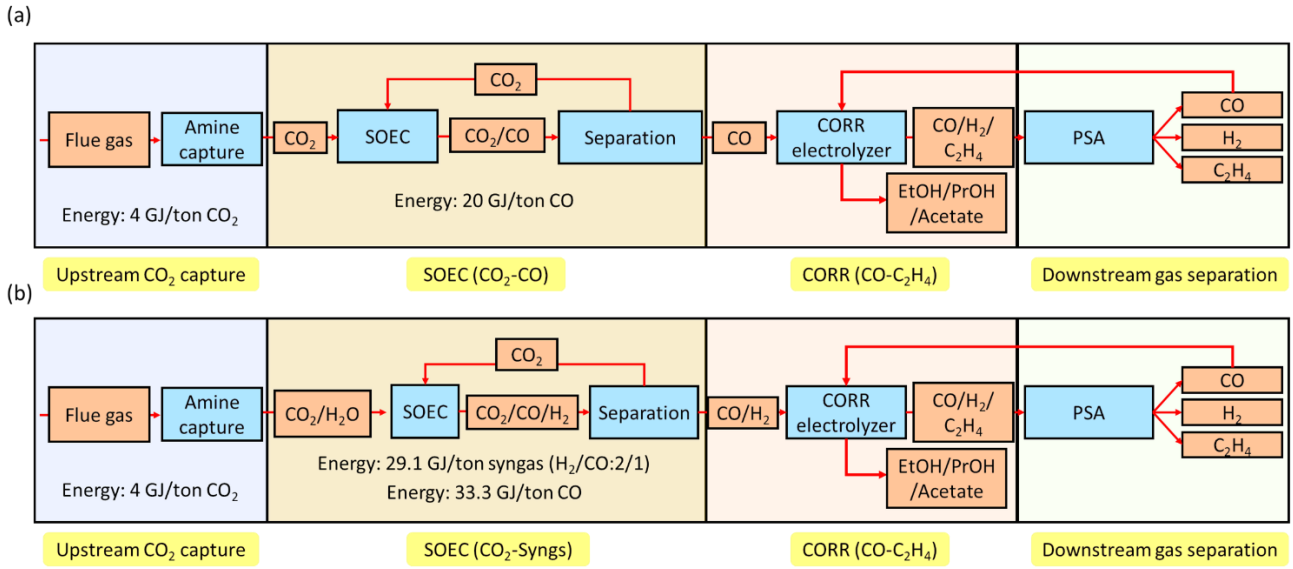
Supplementary Fig. 42 | C_2H_4 FE of bare Cu and Cu/C–COOH for COR from H_2/CO ratio of 2/1 at 2 sccm. Where error bars are shown, values are means, and error bars indicate s.d. (n = 3 replicates).



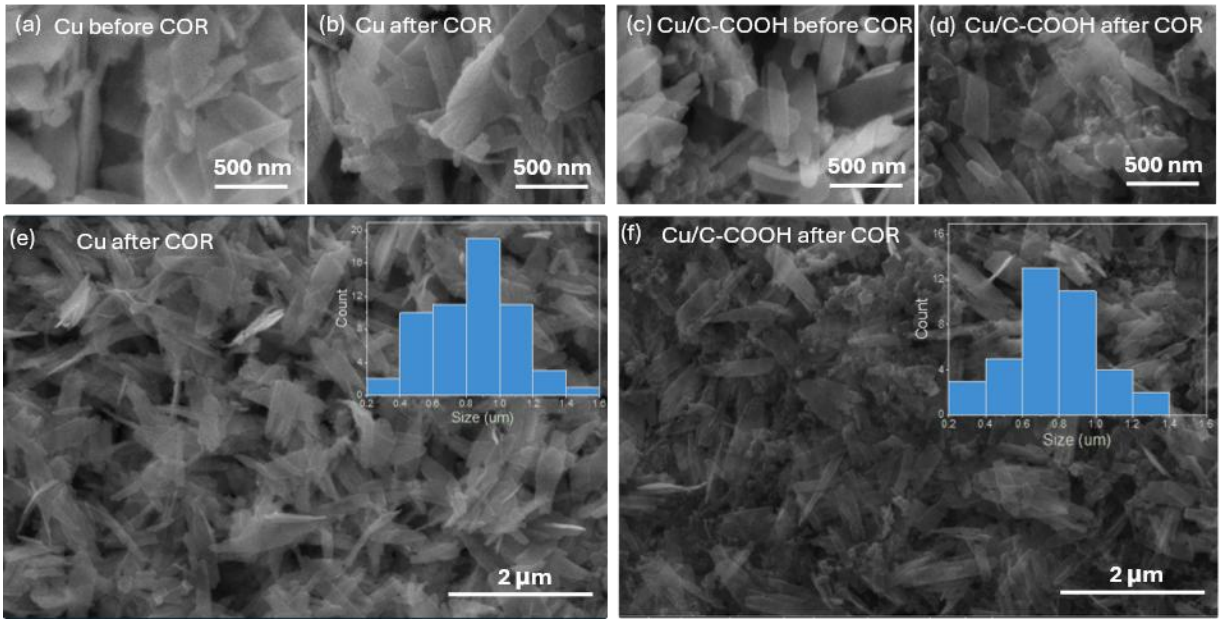
Supplementary Fig. 43 | Impact of local CO concentration on C_2H_4 production rate. FEM simulation of C_2H_4 production rate change with local CO concentration at 200 mA cm^{-2}.



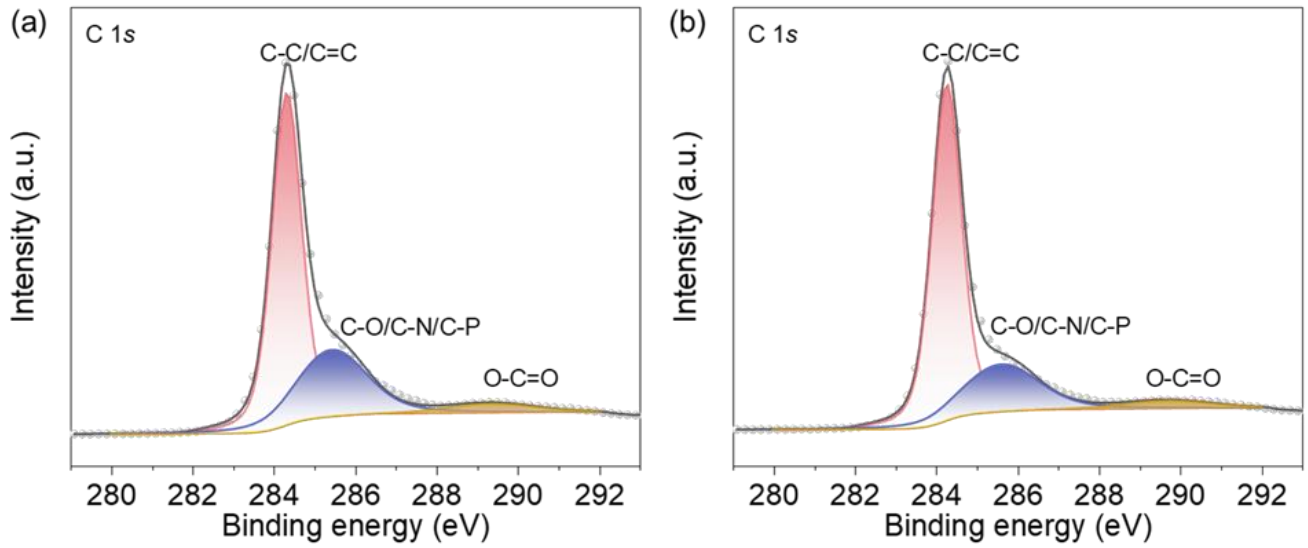
Supplementary Fig. 44 | Ethylene FE of Cu/C–COOH on homemade carbon paper with NiFe-B as anode catalyst for oxygen reduction reaction. Where error bars are shown, values are means, and error bars indicate s.d. (n = 3 replicates).



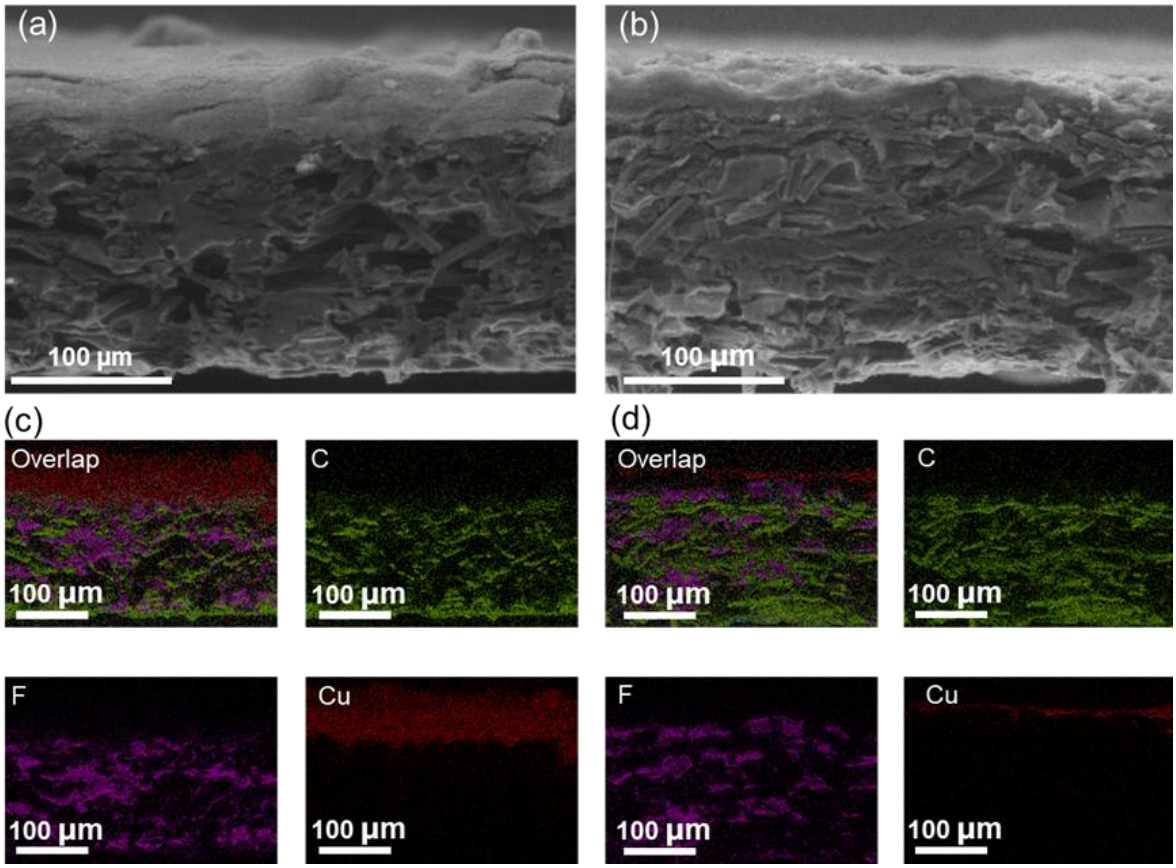
Supplementary Fig. 45 | Flow diagram of the TEA model. The TEA model used for (a) the cascade CO_2-CO-C_2H_4 and (b) CO_2-syngas-C_2H_4 conversion process.



Supplementary Fig. 46 | a,b, Cu electrode (a) before and (b) after COR test. **c,d,** Cu/C–COOH electrode (c) before and (d) after COR test. **e,f,** Particle size distribution on (e) Cu and (f) Cu/C–COOH electrode after COR.



Supplementary Fig. 47 | a,b, Characterization of the functional group on Cu/C–COOH electrode. XPS spectra. XPS C 1s spectra for Cu/C–COOH electrode (a) before and (b) after COR test.



Supplementary Fig. 48 | Characterization of the durability of home-fabricated carbon paper. Cross-section (a, b) SEM images and (c,d) EDS mapping of Cu/C–COOH on home-made carbon paper (a, c) before and (b, d) after COR stability test.

Supplementary Note 1. Multiphysics simulation method

The simulation of local CO distribution in 2D domain were performed in the COMSOL Multiphysics package based on the finite-element-method solver. The local concentration of the ith ionic species $C_i(\boldsymbol{r})$ is described by the Nernst-Plank (NP) equation as follows:[1]

$$\frac{\partial C_i(\boldsymbol{r})}{\partial t} = -\nabla \cdot \boldsymbol{J_i}(r) + \mathrm{R_i}$$
$$= -\nabla \cdot \left(-D_i \nabla C_i(\boldsymbol{r}) - \frac{D_i z_i F C_i(\boldsymbol{r})}{RT} \nabla \phi(\mathbf{r}) \right) + \sum_p R_i \qquad \textbf{(Equation S1)}$$

where $J_i(\boldsymbol{r})$ and D_i refer to the total flux and the diffusion coefficient for the ith ionic species, respectively, R_i is the rate of production of species due to the homogeneous reactions as given by Equation S2, while the other terms are defined the same way as in the PB equation. Magnetic interaction and convection are assumed to be absent in this study. The values of D_i in the electrolyte is given in **Supplementary Table 1**.

$$H_2O \xleftrightarrow{k_{wf},\ k_{wr}} H^+ + OH^- \quad (K_w) \qquad \textbf{(Equation S2)}$$

The heterogeneous reactions S3 and S4 are considered to be occurring at the planar electrode surface in the CO_2RR process:

$$CO\,(aq) + H_2O + 8e^- \rightleftarrows C_2H_4(\mathrm{g}) + 8OH^- \qquad \textbf{(Equation S3)}$$

$$2H_2O + 2e^- \rightleftarrows H_2(\mathrm{g}) + 2OH^- \qquad \textbf{(Equation S4)}$$

The flux of the solution species at the surface is then given by equations (S5)–(S7):

$$\vec{J}_i\big|_{x=0,t} = 0 \quad i = Li^+, H^+, OH^- \qquad \textbf{(Equation S5)}$$

$$\vec{J}_{CO}\Big|_{x=0,t} = \frac{1}{n_{e,C_2H_4}} \times \frac{j_{tot}}{F} \times FE_{C_2H_4} \qquad \textbf{(Equation S6)}$$

$$\vec{J}_{OH^-}\Big|_{x=0,t} = \frac{1}{n_{e,OH^-}} \times \frac{j_{tot}}{F} \qquad \textbf{(Equation S7)}$$

where F is the Faraday's constant, t is the time, n_{e,C_2H_4} is the number of electrons consumed to reduce CO to C_2H_4 and n_{e,OH^-} is the number of electrons consumed per OH^-.

The total current density at the surface is given by:

$$j_{tot}(\eta) = i_0\left\{e^{-\frac{\alpha nF\eta}{RT}} - e^{\frac{(1-\alpha)nF\eta}{RT}}\right\} \qquad \textbf{(Equation S8)}$$

where i_0 is the exchange current, α is the transfer coefficient, n is the number of electrons transferred per half-cell reaction, η is the overpotential. T is the temperature, and F, R are the Faradaic constant and the ideal gas constant, respectively.

The rate of each species R_i, can be broken into carbonate equilibria as follows

$$R_{H^+} = [H_2O]k_{wf} - [H^+][OH^-]\,k_{wr} \qquad \textbf{(Equation S9)}$$

$$R_{OH^-} = [H_2O]k_{wf} - [H^+][OH^-]\,k_{wr} \qquad \textbf{(Equation S10)}$$

The values of the rate constants for reactions S9–S10 are given in **Supplementary Table 2**.

The maximum aqueous CO concentration is assumed to obey Henry's law as given by Equation S11– S12.

$$C_{CO,aq}^{max} = P_{CO_{(g)}}\,K_{Henry,CO}\,10^{-c_{Sechenov,CO}} \qquad \textbf{(Equation S11)}$$

$$K_{Henry,CO} = A_{CO}^{B_{CO}\left(\frac{1}{T}-\frac{1}{298.15[K]}\right)} \qquad \textbf{(Equation S12)}$$

$$c_{Sechenov,CO} = \sum_{i,i\neq CO_{(aq)}} c_i\left[\frac{mol}{dm^3}\right](h_i + h_{G,0} + h_T(T - 198.15[K])) \qquad \textbf{(Equation S13)}$$

in which is the CO partial pressure in the gas phase, A_{CO}, a Henry's law constant for CO, was set to $9.8 \times 10^{-6}\,mol\,m^{-3}\,Pa^{-1}$. B_{CO}, another Henry's law constant for CO, was set to $1300\,K$. h_i is an ion-dependent Sechenov constant, while $h_{G,0}$ and h_T are gas-dependent Sechenov constants, which are provided in **Supplementary Table 3.**

Supplementary Note 2. N_2O titration for measuring the fractional exposure of catalyst surface.

The fractional exposure of the catalyst surface was measured using the N_2O titration method reported by Egbert et al.[2] All experiments were conducted in a small-volume H-type electrochemical cell, using cellulosic filter paper as the membrane (Whatman, 1001–185). The cathode consisted of a 0.5 cm × 0.5 cm piece of home-fabricated carbon paper-supported catalyst, while a 1 cm^2 Pt mesh served as the anode. A saturated KCl-filled Ag/AgCl reference electrode was employed. The electrolyte solution was 20 mM ammonium hexafluorophosphate (NH_4PF_6, Aldrich, >95%), with 16 mL used for both the cathodic and anodic compartments. Electrochemical measurements were carried out using a Metrohm Autolab PGSTAT204 potentiostat equipped with a BOOSTER10A module. The sample pretreatment and titration procedures strictly followed the protocol described by Egbert et al.

Supplementary Note 3. Explanation of the difference between ideal and actual effective activation energies governing the selectivity change during COR.

The selectivity change from 50% to 80% could response to only a few kJ mol^{-1}, which beyond the quantitative accuracy of DFT. Based on the transition state approximation, and assuming that the effective activation energy of the competing reaction does not change, the required decrease in effective activation energy is only about 0.04 eV. In comparison, the average change across all the systems we considered is ~ 0.18 eV on Cu and ~ 0.15 eV on C. However, there are several reasons why the 0.04 eV value may be overly idealized:

1. CO adsorption and coverage effects. CO adsorption is generally very favorable, resulting in a lower standard free energy of adsorbed CO* compared with gas-phase CO. According to the energetic span theory[3], the relevant effective barrier should therefore be defined from CO* to the highest-energy transition state along the pathway. This makes the CO surface coverage a critical factor. While the 0.04 eV estimation would still hold under an ideal Langmuir isotherm, real systems exhibit significant adsorbate–adsorbate interactions[4]. These interactions dampen the increase in CO coverage with adsorption energy, particularly since CO is already strongly adsorbed. Consequently, a larger than 0.04 eV change is required to reproduce the experimental enhancement.

2. DFT error considerations and statistical reliability. We agree that the intrinsic uncertainty of DFT can exceed 0.04 eV. However, our analysis spans graphitic carbon and amorphous carbon systems, and in all cases the results consistently predict stronger adsorption, with a distribution ranging from ~0.1 eV to ~0.3 eV. Statistically, this supports the conclusion that the COOH functional group enhances CO adsorption.

3. The experimentally measured selectivity represents the collective contribution from a distribution of catalytic sites. To achieve an overall increase in selectivity from 50% to 80%, some active sites must exhibit significantly higher intrinsic selectivity (i.e., >80%) to offset less selective regions or nonfunctionalized Cu sites. Therefore, the activation energy difference at the most active or most selective sites must exceed the averaged 0.04 eV value.

Taken together, these theoretical insights provide a rational explanation for the experimentally observed enhancement of CO adsorption on the Cu/C–COOH electrode, as also evidenced by operando Raman spectroscopy (Fig. 3c) and CO adsorption isotherms (Fig. 4b). Our calculations thus complement the operando measurements and offer mechanistic insight into the origin of this effect.

Supplementary Note 4. Porous carbon material engineering for enriching local reactant concentrations via enhanced mass transport.

Under high-SPCE syngas feeds, the primary limitation is H_2 dilution, which lowers the effective interfacial CO activity even when the bulk CO molar feed is adequate. COOH-functionalized carbon creates a near-surface layer that selectively enriches CO at the reaction interface, increasing the local CO activity/coverage relevant for C–C coupling on Cu.

We were initially motivated by previous studies in which porous carbon material engineering and surface modification has been widely employed to enrich local reactant concentrations by facilitating mass transport to the catalytic, thus enhancing reaction selectivity. Similar concepts have been successfully applied in low-concentrated nitric oxide electroreduction to ammonia[5], nitrate electroreduction to ammonia[6], ethylene electrosynthesis from low-concentrated acetylene[7], neutral H_2O_2 electrosynthesis[8], oxygen reduction reaction in PEM fuel cells[9], and CO electroreduction to ethanol[10].

Supplementary Table 1. Diffusion-coefficients in m^2s^{-1}.

Constant	Value	Reference
D_{H^+}	9.31×10^{-9}	[11]
D_{Li^+}	1.03×10^{-9}	[11]
D_{OH^-}	5.27×10^{-9}	[11]
D_{H_2}	4.40×10^{-9}	[12]
D_{CO}	2.03×10^{-9}	[13]

Supplementary Table 2. Rate constants used for modelling.[11]

Constant	Value	Units
k_{wf}	2.4×10^{-2}	$mol\ m^3 s^{-1}$
k_{wr}	2.4×10^{6}	$mol^{-1} m^3 s^{-1}$

Supplementary Table 3. h parameters used to estimate Sechenov's constant in $m^3 kmol^{-1}$. [14]

Constant	Value
h_{Li^+}	0.0922
h_{OH^-}	0.0839
h_{CO}^0	-0.0172
h_{CO}^T	-0.00038

Note: h_{CO}^0 and h_{CO}^T were set to the CO_2 value of $-0.0172\ m^3 kmol^{-1}$ and $-0.00038\ m^3 kmol^{-1}$,respectively, due to the lack of relevant data on CO.

Supplementary Table 4. BET analysis results.

Samples	S_{BET} ($m^2 g^{-1}$)	Total pore volume ($cm^3 g^{-1}$)	Average pore size (nm)
C	68.61	0.15	9.87
C–COOH	80.85	0.16	8.49
C–N	67.33	0.16	9.53
C–P	157.84	0.23	7.04

Supplementary Table 5. Thermocatalytic pathways for the conversion of syngas to ethylene: comparison of the performance with the literature.[15–18]

Reference	Temperature (°C)	Pressure (bar)	Selectivity (%, C_2H_4)	Conversion (%)	Energy intensity (GJ/ton C_2H_4)	Carbon intensity (tCO_2/tC_2H_4)
This work	25	1.0	72	73	203	–2.82*
Fischer-Tropsch[7]	295	5.0	7.0	93.5	306	1.77
Methanol-to-olefins[8]	250	50	41.0	95.0	211	2.01
Direct syngas-to-olefins[9]	350	20	23.7	90.4	46	1.43
Industrial NG-to-olefins (Steam cracking) [18]	400	15			18	2.2

*Assuming a –0.8 ton CO_2/tonne CO_2 captured for the carbon intensity of the CO_2 feedstock[19] and an electricity emission factor of 11 g/kWh$_e$; Downstream production separation considered in this work but not in the thermocatalytic pathways.

Supplementary Table 6. Indexes, filename of files in paper of Deringer et al (ref. 20), number of carbon atoms, and formation energies per atom (E_f) of amorphous carbon structures.

Index	Filename	Number of carbon atoms	E_f (eV)	Index	Filename	Number of carbon atoms	E_f (eV)
1	DFT_relaxed_01_1.xyz	43	0.89	26	DFT_relaxed_06_1.xyz	37	1.01
2	DFT_relaxed_01_2.xyz	47	1.06	**27**	**DFT_relaxed_06_2.xyz**	**43**	**0.71**
3	DFT_relaxed_01_3.xyz	41	1.17	28	DFT_relaxed_06_3.xyz	44	0.89
4	DFT_relaxed_01_4.xyz	40	1.18	29	DFT_relaxed_06_4.xyz	36	1.07
5	DFT_relaxed_01_5.xyz	48	1.07	30	DFT_relaxed_06_5.xyz	42	0.98
6	**DFT_relaxed_02_1.xyz**	**35**	**0.88**	31	DFT_relaxed_07_1.xyz	45	0.91
7	DFT_relaxed_02_2.xyz	47	1.59	32	DFT_relaxed_07_2.xyz	46	1.08
8	DFT_relaxed_02_3.xyz	45	1.16	33	DFT_relaxed_07_3.xyz	50	×
9	DFT_relaxed_02_4.xyz	49	0.94	34	DFT_relaxed_07_4.xyz	47	0.91
10	DFT_relaxed_02_5.xyz	48	×	35	DFT_relaxed_07_5.xyz	41	0.95
11	DFT_relaxed_03_1.xyz	40	1.20	36	DFT_relaxed_08_1.xyz	41	0.96
12	DFT_relaxed_03_2.xyz	50	1.38	37	DFT_relaxed_08_2.xyz	41	0.98
13	DFT_relaxed_03_3.xyz	38	0.96	38	DFT_relaxed_08_3.xyz	45	0.91
14	DFT_relaxed_03_4.xyz	46	1.15	39	DFT_relaxed_08_4.xyz	37	×
15	DFT_relaxed_03_5.xyz	47	0.97	40	DFT_relaxed_08_5.xyz	39	1.05
16	DFT_relaxed_04_1.xyz	48	1.00	41	DFT_relaxed_09_1.xyz	45	1.29
17	DFT_relaxed_04_2.xyz	38	1.05	42	DFT_relaxed_09_2.xyz	43	1.02
18	DFT_relaxed_04_3.xyz	40	1.06	**43**	**DFT_relaxed_09_3.xyz**	**41**	**0.81**
19	**DFT_relaxed_04_4.xyz**	**41**	**0.88**	44	DFT_relaxed_09_4.xyz	43	0.90
20	DFT_relaxed_04_5.xyz	48	0.89	45	DFT_relaxed_09_5.xyz	51	1.08
21	DFT_relaxed_05_1.xyz	44	0.91	46	DFT_relaxed_10_1.xyz	44	×
22	DFT_relaxed_05_2.xyz	48	×	47	DFT_relaxed_10_2.xyz	33	0.93
23	DFT_relaxed_05_3.xyz	37	1.62	48	DFT_relaxed_10_3.xyz	36	1.24
24	DFT_relaxed_05_4.xyz	42	1.17	49	DFT_relaxed_10_4.xyz	45	0.90
25	**DFT_relaxed_05_5.xyz**	**44**	**0.88**	50	DFT_relaxed_10_5.xyz	41	1.10

Note: The filenames highlighted in bold front are the top 5 stable carbon structures. The filenames marked by × are not stable carbon structures.

Supplementary Table 7. Indexes, name of N-functionalized carbon, number of carbon atoms, indexes of the randomly N-functionalized carbon, and formation energies per atom (E_f) of N-functionalized amorphous carbon adopted from file of DFT_relaxed_02_1 in paper of Deringer et al. (ref. 20).

Index	Name of N-functionalized carbon	Number of carbon atoms	Index of N-functionalized carbon atoms	E_f (eV)
1	Amor-C_{02_1}-N-1	35	16,27	-0.12
2	Amor-C_{02_1}-N-2	35	21,1	-0.06
3	Amor-C_{02_1}-N-3	35	20,2	-0.69
4	Amor-C_{02_1}-N-4	35	10,4	-0.42
5	Amor-C_{02_1}-N-5	35	21,1	-0.06
6	Amor-C_{02_1}-N-6	35	6,9	0.00
7	**Amor-C_{02_1}-N-7**	**35**	**10,30**	**-1.62**
8	Amor-C_{02_1}-N-8	35	12,26	-0.26
9	Amor-C_{02_1}-N-9	35	24,9	-0.51
10	Amor-C_{02_1}-N-10	35	5, 34	0.39

Supplementary Table 8. Indexes, name of N-functionalized carbon, number of carbon atoms, indexes of the randomly N-functionalized carbon, and formation energies per atom (E_f) of N-functionalized amorphous carbon adopted from file of DFT_relaxed_04_4 in paper of Deringer et al. (ref. 20).

Index	Name of N-functionalized carbon	Number of carbon atoms	Index of N-functionalized carbon atoms	E_f (eV)
1	Amor-C_{04_4}-N-1	41	1,13	0.05
2	Amor-C_{04_4}-N-2	41	37,33	-0.26
3	Amor-C_{04_4}-N-3	41	7,6	0.01
4	Amor-C_{04_4}-N-4	41	26,31	-0.54
5	**Amor-C_{04_4}-N-5**	**41**	**1,38**	**-0.77**
6	Amor-C_{04_4}-N-6	41	27,29	-0.41
7	Amor-C_{04_4}-N-7	41	32,14	-0.22
8	Amor-C_{04_4}-N-8	41	41, 25	-0.44
9	Amor-C_{04_4}-N-9	41	19, 34	0.25
10	Amor-C_{04_4}-N-10	41	40,38	-0.31

Supplementary Table 9. Indexes, name of N-functionalized carbon, number of carbon atoms, indexes of the randomly N-functionalized carbon, and formation energies per atom (E_f) of N-functionalized amorphous carbon adopted from file of DFT_relaxed_05_5 in paper of Deringer et al. (ref. 20)

Index	Name of N-functionalized carbon	Number of carbon atoms	Index of N-functionalized carbon atoms	E_f (eV)
1	Amor-C_{05_5}-N-1	44	17, 23	-0.53
2	Amor-C_{05_5}-N-2	44	6,18	0.38
3	Amor-C_{05_5}-N-3	44	2,1	0.01
4	Amor-C_{05_5}-N-4	44	31,43	0.55
5	Amor-C_{05_5}-N-5	44	33,44	-0.51
6	Amor-C_{05_5}-N-6	44	42,27	-0.98
7	Amor-C_{05_5}-N-7	44	2,35	-1.14
8	Amor-C_{05_5}-N-8	44	11,34	0.35
9	Amor-C_{05_5}-N-9	44	38,36	0.20
10	**Amor-C_{05_5}-N-10**	**44**	**3,29**	**-1.75**

Supplementary Table 10. Indexes, name of N-functionalized carbon, number of carbon atoms, indexes of the randomly N-functionalized carbon, and formation energies per atom (E_f) of N-functionalized amorphous carbon adopted from file of DFT_relaxed_06_2 in paper of Deringer et al. (ref. 20).

Index	Name of N-functionalized carbon	Number of carbon atoms	Index of N-functionalized carbon atoms	E_f (eV)
1	**Amor-C_{06_2}-N-1**	**43**	**11,3**	**-1.39**
2	Amor-C_{06_2}-N-2	43	19,16	-0.37
3	Amor-C_{06_2}-N-3	43	10,25	0.11
4	Amor-C_{06_2}-N-4	43	9,15	-0.53
5	Amor-C_{06_2}-N-5	43	30,28	0.08
6	Amor-C_{06_2}-N-6	43	27,31	0.40
7	Amor-C_{06_2}-N-7	43	26,43	0.70
8	Amor-C_{06_2}-N-8	43	6,42	0.35
9	Amor-C_{06_2}-N-9	43	10,28	0.16
10	Amor-C_{06_6}-N-10	43	4,7	0.26

Supplementary Table 11. Indexes, name of N-functionalized carbon, number of carbon atoms, indexes of the randomly N-functionalized carbon, and formation energies per atom (E_f) of N-functionalized amorphous carbon adopted from file of DFT_relaxed_09_3 in paper of Deringer et al. (ref. 20).

Index	Name of N-functionalized carbon	Number of carbon atoms	Index of N-functionalized carbon atoms	E_f (eV)
1	Amor-C_{09_3}-N-1	41	19,11	-1.11
2	Amor-C_{09_3}-N-2	41	33, 7	0.57
3	Amor-C_{09_3}-N-3	41	32,4	-0.26
4	**Amor-C_{09_3}-N-4**	**41**	**12,9**	**-1.23**
5	Amor-C_{09_3}-N-5	41	22,10	-0.08
6	Amor-C_{09_3}-N-6	41	21,39	0.46
7	Amor-C_{09_3}-N-7	41	24,27	-0.68
8	Amor-C_{09_3}-N-8	41	15,13	-0.97
9	Amor-C_{09_3}-N-9	41	35,25	-0.99
10	Amor-C_{09_3}-N-10	41	16,3	-0.83

Supplementary Table 12. Indexes, name of P-functionalized carbon, number of carbon atoms, indexes of the randomly P-functionalized carbon, and formation energies per atom (E_f) of P-functionalized amorphous carbon adopted from file of DFT_relaxed_02_1 in paper of Deringer et al. (ref. 20).

Index	Name of P-functionalized carbon	Number of carbon atoms	Index of P-functionalized carbon atoms	E_f (eV)
1	Amor-C_{02_1}-P-1	35	16,27	2.90
2	Amor-C_{02_1}-P-2	35	21,1	-0.67
3	**Amor-C_{02_1}-P-3**	**35**	**20,2**	**-0.78**
4	Amor-C_{02_1}-P-4	35	10,4	-0.52
5	Amor-C_{02_1}-P-5	35	21,1	-0.67
6	Amor-C_{02_1}-P-6	35	6,9	0.29
7	Amor-C_{02_1}-P-7	35	10,30	-0.29
8	Amor-C_{02_1}-P-8	35	12,26	-0.33
9	Amor-C_{02_1}-P-9	35	24,9	0.17
10	Amor-C_{02_1}-P-10	35	5, 34	-0.24

Supplementary Table 13. Indexes, name of P-functionalized carbon, number of carbon atoms, indexes of the randomly P-functionalized carbon, and formation energies per atom (E_f) of P-functionalized amorphous carbon adopted from file of DFT_relaxed_04_4 in paper of Deringer et al. (ref. 20).

Index	Name of P-functionalized carbon	Number of carbon atoms	Index of P-functionalized carbon atoms	E_f (eV)
1	Amor-C_{04_4}-P-1	41	1,13	-0.64
2	Amor-C_{04_4}-P-2	41	37,33	-0.12
3	Amor-C_{04_4}-P-3	41	7,6	-1.27
4	Amor-C_{04_4}-P-4	41	26,31	-1.26
5	Amor-C_{04_4}-P-5	41	1,38	-1.41
6	Amor-C_{04_4}-P-6	41	27,29	-1.57
7	**Amor-C_{04_4}-P-7**	**41**	**32,14**	**-2.11**
8	Amor-C_{04_4}-P-8	41	41, 25	-0.13
9	Amor-C_{04_4}-P-9	41	19, 34	0.11
10	Amor-C_{04_4}-P-10	41	40,38	-1.49

Supplementary Table 14. Indexes, name of P-functionalized carbon, number of carbon atoms, indexes of the randomly P-functionalized carbon, and formation energies per atom (E_f) of P-functionalized amorphous carbon adopted from file of DFT_relaxed_05_5 in paper of Deringer et al. (ref. 20).

Index	Name of P-functionalized carbon	Number of carbon atoms	Index of P-functionalized carbon atoms	E_f (eV)
1	Amor-C_{05_5}-P-1	44	17, 23	-0.52
2	Amor-C_{05_5}-P-2	44	6,18	0.46
3	Amor-C_{05_5}-P-3	44	2,1	0.26
4	**Amor-C_{05_5}-P-4**	**44**	**31,43**	**-1.26**
5	Amor-C_{05_5}-P-5	44	33,44	-0.52
6	Amor-C_{05_5}-P-6	44	42,27	1.79
7	Amor-C_{05_5}-P-7	44	2,35	-0.25
8	Amor-C_{05_5}-P-8	44	11,34	0.43
9	Amor-C_{05_5}-P-9	44	38,36	0.71
10	Amor-C_{05_5}-P-10	44	3,29	-0.43

Supplementary Table 15. Indexes, name of P-functionalized carbon, number of carbon atoms, indexes of the randomly P-functionalized carbon, and formation energies per atom (E_f) of P-functionalized amorphous carbon adopted from file of DFT_relaxed_06_2 in paper of Deringer et al. (ref. 20).

Index	Name of P-functionalized carbon	Number of carbon atoms	Index of P-functionalized carbon atoms	E_f (eV)
1	**Amor-C_{06_2}-P-1**	**43**	**11,3**	**-0.72**
2	Amor-C_{06_2}-P-2	43	19,16	-0.71
3	Amor-C_{06_2}-P-3	43	10,25	-0.04
4	Amor-C_{06_2}-P-4	43	9,15	-0.35
5	Amor-C_{06_2}-P-5	43	30,28	1.44
6	Amor-C_{06_2}-P-6	43	27,31	0.51
7	Amor-C_{06_2}-P-7	43	26,43	1.40
8	Amor-C_{06_2}-P-8	43	6,42	0.40
9	Amor-C_{06_2}-P-9	43	10,28	-0.17
10	Amor-C_{06_6}-P-10	43	4,7	-0.70

Supplementary Table 16. Indexes, name of P-functionalized carbon, number of carbon atoms, indexes of the randomly P-functionalized carbon, and formation energies per atom (E_f) of P-functionalized amorphous carbon adopted from file of DFT_relaxed_09_3 in paper of Deringer et al. (ref. 20).

Index	Name of P-functionalized carbon	Number of carbon atoms	Index of P-functionalized carbon atoms	E_f (eV)
1	Amor-C_{09_3}-P-1	41	19,11	-1.11
2	Amor-C_{09_3}-P-2	41	33, 7	0.57
3	Amor-C_{09_3}-P-3	41	32,4	-0.26
4	**Amor-C_{09_3}-P-4**	**41**	**12,9**	**-1.23**
5	Amor-C_{09_3}-P-5	41	22,10	-0.08
6	Amor-C_{09_3}-P-6	41	21,39	0.46
7	Amor-C_{09_3}-P-7	41	24,27	-0.68
8	Amor-C_{09_3}-P-8	41	15,13	-0.97
9	Amor-C_{09_3}-P-9	41	35,25	-0.99
10	Amor-C_{09_3}-P-10	41	16,3	-0.83

Supplementary Table 17. Indexes, name of COOH-functionalized carbon, number of carbon atoms, indexes of the randomly COOH-functionalized carbon, and formation energies per COOH (E_f) of COOH-functionalized amorphous carbon adopted from file of DFT_relaxed_02_1 in paper of Deringer et al. (ref. 20).

Index	Name of COOH-functionalized carbon	Number of carbon atoms	Index of COOH-functionalized carbon atoms	E_f (eV)
1	Amor-C_{02_1}-COOH-1	35	16,27	-0.47
2	Amor-C_{02_1}-COOH-2	35	21,1	-0.45
3	Amor-C_{02_1}-COOH-3	35	20,2	-0.21
4	**Amor-C_{02_1}-COOH-4**	**35**	**10,4**	**-0.73**
5	Amor-C_{02_1}-COOH-5	35	21,1	-0.44
6	Amor-C_{02_1}-COOH-6	35	6,9	0.56
7	Amor-C_{02_1}-COOH-7	35	10,30	-0.38
8	Amor-C_{02_1}-COOH-8	35	12,26	0.09
9	Amor-C_{02_1}-COOH-9	35	24,9	0.05
10	Amor-C_{02_1}-COOH-10	35	5, 34	0.63

Supplementary Table 18. Indexes, name of COOH-functionalized carbon, number of carbon atoms, indexes of the randomly COOH-functionalized carbon, and formation energies per COOH (E_f) of COOH-functionalized amorphous carbon adopted from file of DFT_relaxed_04_4 in paper of Deringer et al. (ref. 20).

Index	Name of COOH-functionalized carbon	Number of carbon atoms	Index of COOH-functionalized carbon atoms	E_f (eV)
1	Amor-C_{04_4}-COOH-1	41	1,13	-0.48
2	**Amor-C_{04_4}-COOH-2**	**41**	**37,33**	**-1.93**
3	Amor-C_{04_4}-COOH-3	41	7,6	-0.24
4	Amor-C_{04_4}-COOH-4	41	37,31	-0.73
5	Amor-C_{04_4}-COOH-5	41	1,38	-0.67
6	Amor-C_{04_4}-COOH-6	41	27,29	-0.76
7	Amor-C_{04_4}-COOH-7	41	32,14	0.70
8	Amor-C_{04_4}-COOH-8	41	41, 25	-0.28
9	Amor-C_{04_4}-COOH-9	41	19, 34	0.35
10	Amor-C_{04_4}-COOH-10	41	40,38	-0.81

Supplementary Table 19. Indexes, name of COOH-functionalized carbon, number of carbon atoms, indexes of the randomly COOH-functionalized carbon, and formation energies per COOH (E_f) of COOH-functionalized amorphous carbon adopted from file of DFT_relaxed_05_5 in paper of Deringer et al. (ref. 20).

Index	Name of COOH-functionalized carbon	Number of carbon atoms	Index of COOH-functionalized carbon atoms	E_f (eV)
1	Amor-C_{05_5}-COOH-1	44	17, 23	-0.67
2	Amor-C_{05_5}-COOH-2	44	6,18	-0.39
3	Amor-C_{05_5}-COOH-3	44	2,1	-1.32
4	Amor-C_{05_5}-COOH-4	44	31,43	-1.12
5	**Amor-C_{05_5}-COOH-5**	**44**	**33,44**	**-1.43**
6	Amor-C_{05_5}-COOH-6	44	42,27	0.25
7	Amor-C_{05_5}-COOH-7	44	2,35	-0.92
8	Amor-C_{05_5}-COOH-8	44	3,34	1.16
9	Amor-C_{05_5}-COOH-9	44	25,36	1.14
10	Amor-C_{05_5}-COOH-10	44	3,29	0.93

Supplementary Table 20. Indexes, name of COOH-functionalized carbon, number of carbon atoms, indexes of the randomly COOH-functionalized carbon, and formation energies per COOH (E_f) of COOH-functionalized amorphous carbon adopted from file of DFT_relaxed_06_2 in paper of Deringer et al. (ref. 20).

Index	Name of COOH-functionalized carbon	Number of carbon atoms	Index of COOH-functionalized carbon atoms	E_f (eV)
1	Amor-C_{06_2}-COOH-1	43	11,3	-1.28
2	Amor-C_{06_2}-COOH-2	43	19,16	0.00
3	Amor-C_{06_2}-COOH-3	43	24,25	0.03
4	Amor-C_{06_2}-COOH-4	43	9,15	-0.46
5	Amor-C_{06_2}-COOH-5	43	30,28	0.77
6	Amor-C_{06_2}-COOH-6	43	27,31	-0.20
7	Amor-C_{06_2}-COOH-7	43	26,43	0.70
8	Amor-C_{06_2}-COOH-8	43	6,42	1.67
9	Amor-C_{06_2}-COOH-9	43	10,28	-1.02
10	**Amor-C_{06_2}-COOH-10**	**43**	**4,7**	**-1.73**

Supplementary Table 21. Indexes, name of COOH-functionalized carbon, number of carbon atoms, indexes of the randomly COOH-functionalized carbon, and formation energies per COOH (Ef) of COOH-functionalized amorphous carbon adopted from file of DFT_relaxed_09_3 in paper of Deringer et al. (ref. 20).

Index	Name of COOH-functionalized carbon	Number of carbon atoms	Index of COOH-functionalized carbon atoms	E_f (eV)
1	Amor-C_{09_3}-COOH-1	41	19,11	-0.42
2	Amor-C_{09_3}-COOH-2	41	33, 7	-0.12
3	Amor-C_{09_3}-COOH-3	41	32,4	0.08
4	Amor-C_{09_3}-COOH-4	41	12,9	-1.70
5	Amor-C_{09_3}-COOH-5	41	22,10	0.22
6	Amor-C_{09_3}-COOH-6	41	33,39	1.06
7	Amor-C_{09_3}-COOH-7	41	15,27	-1.51
8	Amor-C_{09_3}-COOH-8	41	15,13	-0.73
9	**Amor-C_{09_3}-COOH-9**	**41**	**35,25**	**-1.78**
10	Amor-C_{09_3}-COOH-10	41	16,3	-1.18

Note: the indexes of the randomly COOH-functionalized carbon are slightly different from those used for N- and P-functionalized carbon due to the different functional approaches.

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